Installation Research Projects

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Application Form[1]

Principal Investigator's first and last name
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Applicant (Legal entity)
Faculty of Electrical Engineering, Computer Science and Information Technology Osijek
Full title of the project proposal
Machine Learning for Precision Diagnosis and Robotic Surgery in Oncology
Acronym of the project proposal
ROBOMEDON

The Application Form consists of four parts:

- **Part A - Project proposal (pages 1 - 12)**
- **Part B – Principal investigator (pages 13 - 16)**
- **Part C - Research group and project management**
- **References**

Remarks:

Please note the following text formatting restrictions: font OpenSans or Arial, font size 10, margins (2.0 right and left and 1.5 bottom and top), single line spacing.

It is not allowed to change the Application Form and table templates.

[1] Instructions for completing the Application Form can be found in the Guidelines for Applicants to Croatian Science Foundation's Calls in 2025

Part A - Project Proposal

Part a. State-of-the-art and project proposal

Specify the research question with respect to the current state of the art in the research topic and highlight key publications. Describe your previous research, list major findings and relate them to the project proposal. Explain preliminary results related to the project proposal topic, if there are any. Highlight innovative aspects of the research proposal or multi- or inter- disciplinary aspects. In addition, principal investigators who have been involved in HRZZ projects as principal investigators/collaborators should briefly explain their role in the project, the objectives of the project, the obtained results and how they relate to the proposed research, and provide a link to the project.

Early diagnosis of prostate cancer, breast cancer, and dermatological disorders is crucial for successful treatment and mortality reduction. Detecting these diseases at an early stage enables timely initiation of therapy, significantly increasing the chances of recovery and improving patients' quality of life [1]. Prostate cancer is the most common form of cancer among men in Croatia. According to the Croatian Institute of Public Health (HZJZ), 2,609 new cases of prostate cancer were recorded in 2022, accounting for 19% of all cancer diagnoses in men that year [1]. Globally, prostate cancer also represents a significant public health challenge, with high incidence rates reported in many countries [2]. Breast cancer is the most frequently diagnosed cancer among women in Croatia. In 2022, 3,088 new cases of breast cancer were reported, representing 25% of all cancer diagnoses in women that year [1]. On a global scale, breast cancer remains the leading cause of morbidity and mortality among women, with millions of new cases diagnosed annually. Dermatological disorders, particularly malignant melanoma, pose a serious health threat [3]. Despite advances in diagnostics and treatment, Croatia continues to record a high cancer mortality rate [1]. These statistics underscore the urgent need for improved preventive measures, early diagnosis, and effective treatment strategies to reduce cancer-related mortality [4]. **Early diagnosis facilitates better treatment outcomes, reduces healthcare costs, and enhances patients' quality of life.** Therefore, investing in education, preventive screenings, and advanced diagnostic technologies is of paramount importance to ensure early detection of these diseases [5].

Current State in the Field of Machine Learning Applications in Histological Image Analysis

The development of digital pathology and the application of artificial intelligence (AI) have significantly advanced the diagnosis and analysis of histological samples, particularly in oncology [6] [7] [8]. Traditional analysis methods based on hematoxylin-eosin (H&E) staining and immunohistochemical (IHC) techniques require highly trained pathologists, are often time-consuming, and are limited by sample availability [9]. The introduction of whole-slide imaging (WSI) scanners [10] has enabled the digitization of histological specimens, paving the way for computational image processing and automated analysis methods [11]. However, variations in scanner technical specifications and digitization methods can affect data quality and pose challenges when applying AI models for analysis [12]. Recent research demonstrates that deep learning models, including transformer architectures [13] [14], can reliably segment tumor lesions, identify biomarkers, and predict clinical outcomes based on histological images [15]. The application of generative models, such as the VirtualMultiplexer system [16], facilitates virtual tissue staining and the generation of multi-channel data without requiring multiple staining procedures, significantly accelerating pathologists' workflows [17]. Despite these advancements, challenges remain in data standardization, AI model interpretability, and integration into clinical workflows [18]. Differences between WSI scanners, variations in staining and morphology of samples, and the need for large annotated datasets present obstacles to broader adoption [19] [20] [21]. Additionally, ethical and regulatory concerns, including data privacy and accountability for AI-based diagnostic decisions, are critical aspects that must be addressed [22]. ***Further progress in this field depends on the development of robust, transparent, and generalizable AI models, the integration of multimodal data (genomic, histological, and clinical data), and enhanced collaboration between computer scientists, pathologists, and clinicians to ensure widespread application in diagnostics and personalized medicine.***

Fusion Biopsy of the Prostate

The current state of advanced machine learning methods for precise fusion biopsy of the prostate based on multiparametric MRI and ultrasound demonstrates significant progress in integrating

artificial intelligence (AI) into prostate cancer diagnostics [23]. Multiparametric MRI (mpMRI) has become the gold standard [24] for detecting and characterizing tumor lesions, particularly for highly suspicious lesions categorized as PIRADS 4 and PIRADS 5 [25] [26]. However, biopsy precision still largely depends on the expertise of radiologists and urologists, leading to variability in diagnostics. Research trends are increasingly focused on automating key steps in the fusion biopsy process, including prostate and lesion segmentation, MRI-ultrasound registration, and 3D visualization of tumor regions [27] [28]. The application of deep neural networks enables precise segmentation of the prostate and suspicious lesions, while attention-based architectures (ViT, Swin Transformer) [29] enhance MRI image interpretation and improve tumor boundary delineation [30]. Additionally, deformable registration methods [31], such as VoxelMorph [32], and graph neural networks (GNNs) [33] facilitate better spatial alignment of MRI and ultrasound data, improving biopsy navigation accuracy [34]. Advanced self-supervised learning approaches enhance the adaptability of AI models to different datasets without requiring large amounts of manually annotated images [35] [36]. One of the key challenges in implementing AI systems in clinical practice is model interpretability. Consequently, explainable AI (XAI) methods are being increasingly explored to help clinicians understand model decisions and build trust in its application [37]. Moreover, volumetric 3D reconstruction of the prostate and tumor lesions significantly improves biopsy navigation and enhances therapy planning [38]. ***The integration of deep neural networks, self-supervised learning, and deformable registration is expected to greatly enhance the precision of fusion biopsy based on multiparametric MRI and ultrasound, reducing inter-physician variability and increasing the detection of clinically significant tumor lesions.*** Further advancements in this field are anticipated to result in highly accurate, standardized, and objective fusion biopsy systems, significantly improving diagnostics and enabling a more personalized approach to prostate cancer treatment.

Robot-Assisted Radical Prostatectomy

Current research in the field of intraoperative detection and preservation of neurovascular structures during robot-assisted radical prostatectomy (RARP) is based on the integration of advanced imaging techniques, surgical navigation, and computational data analysis [39] [40]. Robotic systems such as the Da Vinci platform [41] [42] [43] [44] enable highly precise surgical procedures with enhanced visualization of the operative field. However, surgical decision-making and the detection of critical anatomical structures still largely depend on the surgeon's subjective assessment [45] [46]. Key challenges include the lack of standardized methods for intraoperative identification of neurovascular structures, anatomical variability among patients, and the limitations of real-time imaging modalities [47]. Conventional approaches, such as fluorescence angiography and intraoperative ultrasound, provide valuable information but are often insufficiently precise for detecting fine neurovascular structures that are crucial for preserving functional outcomes in patients [48]. ***The proposed research suggests the application of hyperspectral imaging in robotic surgery***, which enables non-invasive real-time tissue visualization based on spectral characterization [49]. Hyperspectral cameras generate a rich spectrum of data that can differentiate between tissue types, detect neurovascular structures with high precision, and potentially reduce intraoperative complications [50] [51]. Furthermore, the development of deep neural networks (CNNs, Transformer-based models) for automated segmentation and classification of neurovascular structures represents a significant advancement in this field [52]. Additionally, a strong focus is placed on the integration of explainable AI (XAI) methods, which enhance model interpretability and increase surgeons' trust in AI-driven systems [53]. The application of these methods holds great potential for improving intraoperative decision-making, reducing surgical errors, and enhancing postoperative functional outcomes. Consequently, robotic prostate surgery is expected to reach new levels of precision and safety.

Processing and Analysis of Intraoperative Videos in Robotic Surgery

Current research in the field of intraoperative video analysis in robotic surgery is focused on the application of computer vision and deep learning for the automatic segmentation of surgical instruments, identification of anatomical structures, and recognition of key surgical phases [54] [55] [56] [57]. Robotic systems such as the Da Vinci platform [58] [59] [60] enable high-resolution recording of the operative field. However, the analysis of these videos still heavily relies on manual data annotation and the subjective assessment of surgeons [61].

Previous studies have explored deep neural network-based methods (CNNs, RNNs) for instrument detection [62], but their application remains limited due to variations in lighting conditions, different surgical techniques, and the complexity of the intraoperative environment. Additionally, ***most existing solutions focus on postoperative analysis, while real-time analysis methods are still in the early stages of development.*** Key challenges include the lack of large annotated datasets, the need for computationally efficient real-time methods, and the interpretability of models to ensure that surgeons can understand and utilize AI recommendations during surgery [63]. Moreover, current approaches often do not integrate multiple data modalities, such as combining intraoperative videos with surgical records and robotic system sensor data, which could enhance the analysis of surgical workflows [64]. To address these challenges, the proposed research integrates Transformer-based architectures, such as Video Swin-Transformer and TimeSformer, for deep video analysis, along with Graph Neural Networks (GNNs) to model relationships between instruments, tissues, and surgical actions [65]. The implementation of this intelligent system in robotic surgery will enable real-time surgical procedure analysis, improve precision and safety, reduce intraoperative complications, and optimize surgical protocols, significantly enhancing the quality of surgical care.

Integration of Artificial Intelligence into Clinical Practice

The integration of artificial intelligence (AI) into clinical practice faces numerous challenges that hinder its widespread adoption, despite significant advancements in machine learning methods and deep neural networks [66]. One of the primary obstacles is the limited availability of high-quality, standardized, and annotated medical data, which complicates the training of reliable models capable of generalizing across diverse clinical scenarios [67]. Additionally, ***the lack of interpretability in AI*** decision-making impedes its acceptance by healthcare professionals, as most current models function as "black-box" systems without clear explanations of how they arrive at specific diagnostic or therapeutic recommendations [68]. To address these challenges, a key focus of this research will be the validation of developed AI methods through collaboration with medical professionals, involving the testing of AI models on real clinical data and the evaluation of their recommendations against expert assessments [69]. AI systems will be integrated into hospital electronic health records (EHR) via standardized API solutions, enabling their real-time application in clinical decision-making [70]. This approach not only enhances diagnostic accuracy and optimizes surgical procedures but also ensures the safe, scalable, and transparent implementation of AI in medical practice, thereby improving clinical applications and patient care. ***The project involves interdisciplinary collaboration between experts in computer science, medicine, biotechnology, and healthcare informatics***, fostering the development of a system that is both technically innovative and aligned with real clinical needs and regulatory requirements.

Previously conducted research

During the successful implementation and completion of the establishment-research project *"Methods for Interpreting Medical Images for Detailed Analysis of Heart Health – IMAGINEHEART" (UIP-05-2017)*, led by Prof. Dr. Sc. Irena Galić, new research perspectives in medical image processing and analysis were opened. As a PhD candidate and collaborator on the IMAGINEHEART project, I actively contributed to the development and application of deep learning methods for segmentation and analysis of cardiovascular medical images, resulting in significant scientific contributions. The primary goal of the project was to develop methods that, using a minimal number of medical images, enable the quantification of a wide range of heart parameters and their integration into a comprehensive assessment of cardiovascular health. ***The project results were published in 9 scientific papers in Web of Science-indexed journals (Q1, Q2), while 14 scientific papers were presented at peer-reviewed international conferences.*** As part of the project, two PhD candidates were enrolled in a joint doctoral program with Ghent University, where I successfully defended my doctoral dissertation. During the project, Dr. Sc. Marin Benčević also successfully defended his dual PhD dissertation. Additionally, three master's students from FERIT Osijek participated in the project, completing their theses on IMAGINEHEART-related topics. Throughout the project, we developed advanced segmentation methods for the cardiovascular system based on deep learning, including: methods for whole-heart segmentation and segmentation of its individual components from CT and MRI images, methods for detection and tracking of coronary arteries in X-ray angiographic images, methods for segmenting the

left atrial appendage, algorithms for segmenting elliptical structures in medical images, including lesions and epicardial fat. The developed algorithms achieved results comparable to or superior to state-of-the-art methods. The source code, developed libraries, and detailed project information are available to the research community through the IMAGINEHEART platform, while the complete list of publications is accessible on CroRIS. Through this project, significant expertise was gained in 2D and 3D processing of cardiovascular medical images, as well as extensive knowledge in the development of machine learning and deep learning algorithms.

Preliminary Research

In our previous research, we developed state-of-the-art deep learning methods for organ segmentation, including the heart and vascular system. This work serves as a crucial foundation for further studies, as it enables precise anatomical structure analysis and the advancement of AI models that facilitate innovative diagnostic and therapeutic approaches. These include fusion analysis of multimodal medical imaging, intraoperative navigation, and automated quantification of pathological changes.

Part b. Methodology

Describe the research methodology. Specify and describe in detail planned analyses and procedures, and novelty of methodological approaches, the equipment and software licences, the number and charcteristics of samples/study participants (e.g. use of existing or new banks/databases, sampling). Explain why the planned methodology is best suited to achieve the objectives of the project.

The proposed project consists of **five work packages**, each corresponding to a specific thematic group outlined in the project objectives. The following sections detail the **methodology** for each proposed work package.

WP1: Development of Methods for Processing and Analysis of Histopathological Images

The proposed work package focuses on integrating whole-slide imaging (WSI) high-resolution images from the Histolog scanner [71] with innovative machine learning methods ***to enhance the detection and classification of pathological changes in breast, prostate, and dermatological tissue samples.*** Based on the hypothesis that combining advanced machine learning models with optimized processing techniques for extremely large histological images (minimum size of 2GB per image) enables more precise biomarker detection and interpretable tumor microenvironment analysis, this work plan will address key technical challenges related to processing large-scale digital pathology images, segmentation and classification of histological structures, and the application of multimodal data (***genomic, histological, and clinical data***).

Within this work package, activities will include collecting high-resolution histological images (WSI) from the Histolog scanner at IMC Priora hospital, organizing data into a structured database suitable for machine learning, annotating images in collaboration with pathologists to define regions of interest (ROIs), and establishing standardized image acquisition protocols to reduce variations between samples. Preprocessing will involve developing new tiling methods and multi-resolution analysis to divide images into processable segments without information loss. The development and optimization of machine learning models will focus on tissue segmentation and detection of pathological changes, utilizing transformer and diffusion models, as well as unsupervised, self-supervised, and active learning methods to improve histological structure analysis. Tumor subtype classification will be performed by implementing deep convolutional networks such as EfficientNet, ResNet, and DenseNet, combined with attention-based models to enhance classification interpretability and accuracy. Multimodal analysis will include the development of models that combine histological images with genetic and clinical data and ***the exploration of Graph Neural Networks (GNNs) to model the relationships between tumor regions and their microenvironment***. The application of self-supervised learning will improve the model's ability to recognize relevant structures even with limited labeled data, while Graph Neural Networks (GNNs) will enable the analysis of spatial relationships between tumor cells and their microenvironment. ***A significant contribution will be the use of generative models for virtual staining, allowing the synthesis of multichannel data from standard H&E images, thus reducing the need for multiple immunohistochemical stains***. Special attention will be given to developing interpretable AI models (XAI) to ensure the transparency of predictions and their clinical

applicability. The expected outcomes of this project include improved automation of diagnostic processes, enhanced consistency in histological sample analysis, and more precise and faster clinical decision-making in oncology.

WP2: Development of Advanced Machine Learning Methods for Precise Fusion Biopsy of the Prostate Based on Multiparametric MRI and Ultrasound

This work package focuses on the development and optimization of machine learning methods for precise fusion biopsy of the prostate to improve the accuracy of tumor lesion detection. The planned research includes automated segmentation of suspicious PIRADS 4 and PIRADS 5 lesions, multimodal registration of MRI and ultrasound data, and 3D visualization of the prostate and tumor lesions to enhance biopsy precision. ***The goal is to reduce reliance on physicians' cognitive abilities and enable a standardized, objective, and improved approach to prostate biopsy.*** Data collection will involve acquiring new clinical datasets from IMC Priora Hospital.

The planned analyses include contrast and intensity normalization of MRI and ultrasound images, along with intermodal registration to ensure input data consistency. The development of AI models will involve deep learning methods for segmentation and the application of attention-based architectures such as ViT and Swin Transformer. A multimodal registration algorithm, including VoxelMorph deformable networks and Graph Neural Networks (GNNs), will be developed to achieve precise mapping between MRI and ultrasound images. Additionally, personalized 3D models of the prostate will be created based on MRI data to enhance biopsy navigation, while volumetric reconstruction will provide real-time lesion visualization. The innovative aspects of this approach include automatic AI-based lesion segmentation, deep learning for deformable MRI-ultrasound registration, interactive 3D prostate visualization, and multimodal analysis through GNNs. The proposed methodology combines deep learning segmentation, image registration, 3D reconstruction, and multimodal analysis to establish a highly precise system for fusion biopsy of the prostate. The use of unsupervised, self-supervised, and active learning methods will enable better adaptation to different data sources, while explainable AI (XAI) techniques will ensure model interpretability and greater reliability in clinical applications. ***This approach is expected to significantly improve prostate cancer detection, reduce unnecessary biopsies, and enhance diagnostic and treatment accuracy for patients.***

WP3: Development of an AI-Assisted System for Intraoperative Detection and Preservation of Neurovascular Structures During Robot-Assisted Radical Prostatectomy Using Hyperspectral Analysis

WP3 focuses on developing an artificial intelligence-based system utilizing hyperspectral analysis for intraoperative detection of neurovascular structures and tumor tissue during Da Vinci robot-assisted radical prostatectomy. ***The aim of the research is to enable more precise real-time visualization of key anatomical structures, thereby reducing the risk of postoperative complications such as urinary incontinence and erectile dysfunction.*** The methodology includes collecting hyperspectral data during surgeries, developing AI models for tissue segmentation and classification, and evaluating these models through histological validation of segmentation accuracy. Data collection will take place at the IMC Priora clinical center, where hyperspectral imaging will be performed during prostatectomy using a hyperspectral camera integrated into the Da Vinci system. The acquired data will be stored in a structured database, followed by preprocessing steps such as spectral normalization to minimize the effects of lighting variations and the registration of hyperspectral data with the intraoperative video feed from the Da Vinci system. The development of AI models will involve tissue segmentation using deep neural networks, while Vision Transformers (ViT) and Swin Transformer architectures will be employed for enhanced spectral feature analysis. Tissue classification will be based on multimodal models that integrate hyperspectral data with intraoperative RGB images, while Graph Neural Networks (GNNs) will be used to model the relationships between healthy and tumor tissues. Furthermore, Explainable AI (XAI) methods such as Grad-CAM and SHAP will be applied to improve model interpretability and facilitate better interaction between the AI system and surgeons. Model evaluation will be conducted by comparing AI predictions with histological analyses of postoperative samples to ensure high system accuracy.

WP4: Development of an Intelligent System for Real-Time Analysis of Intraoperative Video in Robotic Surgery

The goal of this research is to develop an AI-assisted system for analyzing intraoperative videos in robotic surgery, focusing on the recognition of surgical instruments, automatic annotation of critical surgical steps, and detection of anomalies during surgical procedures, specifically prostatectomies. Robotic surgical systems such as the Da Vinci system enable the recording of high-resolution videos during operations, but there is currently no automated and standardized method for analyzing this data to optimize surgical procedures. By developing new machine learning and deep learning methods for video processing, it will be possible to track surgical instruments, automatically identify anatomical structures and critical phases of the operation, and detect intraoperative complications in real time. ***This work package focuses on the development of an AI-supported system for intraoperative video analysis in robotic surgery, aiming for the automatic detection of surgical instruments, identification of anatomical structures, and recognition of key steps in prostatectomy.*** Data will be collected from existing robotic surgery datasets, including Cholec80, EndoVis Challenges, and publicly available Da Vinci datasets, while additional videos will be gathered in collaboration with clinical partners to train models on real surgical procedures.

Video preprocessing will include lighting normalization, image stabilization, keyframe segmentation, and data registration to optimize AI models. The development of AI models for intraoperative video analysis will be based on advanced deep learning techniques. Models for detecting and tracking surgical instruments will be developed using Transformer-based video analysis methods such as TimeSformer, Video Swin-Transformer, and SlowFast Networks. Additionally, deep neural networks (CNNs) will be implemented to identify anatomical structures and localize them precisely during surgery. Multimodal AI models will be developed to combine video data with intraoperative notes, enabling automatic identification of surgical steps. Graph Neural Networks (GNNs) will be used to model relationships between instruments, tissues, and surgical actions, improving the understanding of intraoperative context. The proposed approach integrates deep learning techniques for video processing, transformer architectures for analyzing surgical workflows, and multimodal AI systems that combine video data with surgical annotations. The application of self-supervised learning methods will improve the generalization of models, while explainable AI (XAI) approaches will ensure transparency and comprehensibility for surgeons. ***This approach will enable the development of an AI-assisted intraoperative system that provides surgeons with real-time automatic recommendations and alerts, significantly enhancing patient safety and the precision of robotic surgical procedures.***

WP5: Advancing the Integration of Artificial Intelligence into Clinical Practice

This work package focuses on developing advanced AI methods to enable more efficient processing, analysis, and interpretation of medical data, as well as their adaptation to specific diagnostic and surgical protocols. The objective of this work package is to facilitate the transparent and reliable integration of AI systems into daily clinical practice by optimizing machine learning methods, developing interpretable models, and creating standardized protocols for their integration into medical information systems. The implementation of explainable AI (XAI) methods will ensure model interpretability, while validation will be conducted through clinical studies and integration into hospital systems. Close collaboration with medical professionals will enable the collection of high-quality feedback on the performance of the developed AI methods, increasing the confidence of healthcare personnel in AI-based systems.

Expected outcomes include the development of scalable, modular, and reliable AI systems, improved diagnostic accuracy, optimization of surgical procedures through real-time intraoperative data analysis, and the safe and ethically compliant implementation of AI in clinical practice, contributing to significant advancements in personalized medicine and oncological surgery. This work package represents a synergy of the first four work packages, bridging the gap between research and the real-world clinical application of the developed AI methods.

An additional challenge is the regulatory framework surrounding the use of artificial intelligence in medicine. In line with new European regulations such as the AI Act and GDPR, it is crucial to ensure

that the developed AI models meet stringent requirements regarding safety, patient privacy, and ethical data usage. Special emphasis will be placed on data anonymization and protection, as well as on ensuring fairness in AI models to prevent potential biases that could lead to unequal medical outcomes across different patient groups. ***This work package integrates the outcomes of the preceding work packages and facilitates the practical implementation of AI methods in real clinical settings.***

Part c. Work plan

In this part of the project proposal it should be clear how the proposed research objectives of the project will be achieved (list the research objectives and planned results).

c1. Research objectives of the project *(list the research objectives of the proposed project)*

O1 Improvement of cancer diagnostics through automated processing and analysis of histopathological images.

O2 Optimization of the fusion biopsy procedure for the prostate based on multiparametric MRI and ultrasound through the development of machine learning methods

O3 Development of methods for precise intraoperative detection and preservation of neurovascular structures during robot-assisted radical prostatectomy using hyperspectral analysis

O4 Implementation of an intelligent system for the automatic analysis of intraoperative video in robotic surgery

O5 Advancement of artificial intelligence integration into clinical practice through the development of advanced models for medical data analysis and their adaptation to specific diagnostic and surgical protocols

c2. Expected results and impact on the research area and beyond

For each research objective, describe planned results and pathways to their achievement. Describe the expected scientific impact (e.g. new scientific knowledge, methodologies, publishing or archiving of scientific articles or data sets) as well as social or economic impact. Given the expected results of the research, indicate who would be users of the research results and the expected impact of the research if applied.

The ROBOMEDON project will be carried out over a five-year period, with clearly defined phases for research, development, and implementation of artificial intelligence methods for medical diagnostics and robotic surgery.

Main Project Outcomes:

1. Development of a software solution for processing and analyzing histopathological images using machine learning to improve oncology diagnostics (linked to objective O1).
2. Development of machine learning methods for precise fusion biopsy of the prostate based on multiparametric MRI and ultrasound, improving the accuracy of tumor lesion detection (linked to objective O2).
3. Development of an AI system for intraoperative detection of neurovascular structures during robot-assisted radical prostatectomy using hyperspectral analysis, enhancing surgical safety and outcomes (linked to objective O3).
4. Development of an AI-assisted system for the automatic analysis of intraoperative robotic surgery videos to optimize surgical procedures (linked to objective O4).
5. Advancement of AI integration into clinical practice through the development of interpretable models for medical data analysis and adaptation to specific diagnostic and surgical protocols (linked to objective O5).

<u>The first year</u> will focus on collecting data necessary for method development (C1, C2, C3). Technical reports will be produced on existing machine learning methods for histopathological image analysis (C1), MRI-ultrasound fusion (C2), and intraoperative hyperspectral analysis (C3). Collaboration with hospital centers will be established to collect relevant clinical data. Special attention will be given to developing a method for processing and analyzing histopathological images (linked to C1, result R1.3.), which will be published at an international scientific conference (result R1.4). Efforts will also focus on developing initial versions of methods for segmenting suspicious prostate lesions (linked to C2, result R1.5) and MRI-ultrasound image registration (linked to C2, result R1.6). Expected scientific output

includes the publication of initial findings in technical reports and preparation for scientific publications.

The second year will focus on developing algorithms and methods for automatic analysis of medical images and videos (objectives C1, C2, C4). Methods for multimodal registration of MRI and ultrasound data will be developed, along with the publication of a scientific paper in a Q1/Q2 journal (C2, results R2.1, R2.2.), and the segmentation of neurovascular structures and tumor tissue (linked to C3, result R2.3). Participation in an international scientific conference is planned with a developed method for detecting and tracking surgical instruments (linked to C4, result R2.4). The collection of clinical data will continue in collaboration with IMC Priora (linked to C2, result R2.5). Additionally, a method for predicting tumor aggressiveness based on multiparametric MRI data will be developed and validated using publicly available datasets (linked to C2, result R2.6).

The third year will focus on developing a method for fusion biopsy of the prostate and presenting findings at an international scientific conference (linked to C2, result R3.2). Research will also focus on creating 3D models of the prostate for visualizing tumor lesions (linked to C2, result R3.3). Studies on integrating a hyperspectral camera into the Da Vinci platform will be conducted, along with the development and testing of advanced hyperspectral analysis models in intraoperative settings (linked to C3, result R3.5), in collaboration with medical experts. A special focus will be placed on integrating methods to improve model interpretability (linked to C5, result R3.4). A Q1/Q2 journal publication will be prepared on the alignment of hyperspectral and intraoperative video data (linked to C3, result R3.5). Two international conference presentations are planned (linked to C2, result R3.2, and C4, result R3.7). The first prototypes of the software frameworks ROBOMED-HistoAI (C1), ROBOMED-Fusion (C2), and ROBOMED-VideoAI (C4) will be developed, with clinical validation and real-world testing to assess model performance.

The fourth year will involve the development of a method for lesion reconstruction to assist during biopsies, followed by a conference presentation (linked to C2, result R4.1). Another planned conference will focus on the developed method for intraoperative visualization of segmented tissues (linked to C3, result R4.2). Further research on explainable AI approaches will be conducted in collaboration with clinicians to ensure the precision of developed methods (linked to C3, result R4.3). A Q1/Q2 journal publication will focus on a method for modeling relationships between surgical instruments, tissues, and surgical actions (linked to C4, result R4.4). Additionally, a method for tracking intraoperative instrument movements and calculating surgical precision will be developed (linked to C4, result R4.6).

The fifth year will be dedicated to the final evaluation and validation of developed solutions in clinical systems (C1, C2, C3, C4, C5). Final technical reports will be completed, and key findings will be published in scientific journals, with an emphasis on clinical applications. The software frameworks integrating the methods developed over the previous four years—including ROBOMED-HistoAI (C1), ROBOMED-Fusion (C2), ROBOMED-HyperSpectral (C3), ROBOMED-VideoAI (C4), and ROBOMED-XAI (C5)—will be finalized and ready for implementation. Special attention will be given to model interpretability through the development of ROBOMED-XAI, a modular system for understanding AI decisions (C5). A training workshop will be held for medical professionals to increase the acceptance of AI technology in clinical practice.

The expected impact of this project includes the development of novel scientific methods, improvements in diagnostic precision, and enhanced surgical safety. The results will be beneficial to medical professionals, AI researchers, hospitals, and technology companies developing advanced AI solutions for medical practice. In addition to scientific publications, each work package will result in software implementations of the developed methods and an experimental analysis of the software's accuracy and applicability. **The data and the specific amount of images planned to be collected are described in detail in the Data Management Plan.**

Research Impact

Given the number of collaborators involved in the project, the goal is to publish 4 Q1/Q2 journal papers (Web of Science) and present intermediate results at 7 relevant international scientific conferences. Additionally, the datasets of histopathological, ultrasound, and MRI images generated during the project will be made publicly available, supporting the research community in the

development and evaluation of new medical image analysis methods. ***This initiative will have a strong impact on the research community and contribute to future studies in the field.***

Part d. Establishing new research group

Describe in detail the process and dynamics of setting up the research group/team and the connection of the proposed research with previous research of the team members. Describe other research implemented at the Organization that might be relevant for evaluating this project proposal.

As part of this project, a ***new interdisciplinary research group*** will be formed to focus on the application of artificial intelligence in medical diagnostics and robotic surgery. The establishment of this group involves connecting researchers with expertise in computer science, medical image analysis, mathematical modeling, and biomedical engineering. The group will include scientists from various institutions, ***including FERIT Osijek*** (Dr. Marija Habijan, Assist. Prof. Hrvoje Leventić, Dr. Marin Benčević), ***Ghent University*** (Dr. Srđan Lazendić), ***IMC Priora*** (Dr. Marija Habijan, Danijel Marinčić, MD), ***General Hospital Dr. Josip Benčević in Slavonski Brod*** (Assist. Prof. Josip Samardžić, Danijel Marinčić, MD, Branimir Bošnjak, MD), and ***University of Slavonski Brod*** (Petar Nakić, MSc. Comp. Eng., Maja Čuletić Čondrić). ***The group's formation will follow a phased integration, with the core team established in the first year and additional researchers, particularly PhD candidates, joining in later stages.***

Previous research conducted by the team includes the development of 2D and 3D methods for cardiovascular image processing and analysis (CT, MRI, angiography), such as whole-heart segmentation and segmentation of its individual components from CT and MRI images, detection and tracking of coronary arteries in X-ray angiography images, segmentation of the left atrial appendage, new algorithms for segmenting elliptical structures in medical images, including lesions and epicardial fat. This expertise forms a solid foundation for the proposed project. Additionally, existing collaborations with medical centers will facilitate testing and increase the likelihood of clinical application of the developed methods. ***The research topic is closely aligned with the group's previous studies in medical image processing and machine learning.***

At FERIT Osijek, ongoing research focuses on machine learning, artificial intelligence, and medical image analysis, all of which are crucial for the development of diagnostic and therapeutic methods in medicine. Through various scientific research projects, FERIT has actively contributed to the development of advanced algorithms for digital medical image processing, anomaly detection in medical data, and the integration of different medical imaging modalities to create more precise diagnostic solutions. Key areas of research include: computational methods for EEG signal analysis, development of sophisticated computational models for cardiovascular health assessment, application of machine learning for predicting treatment outcomes. These advanced algorithms enable better disease detection and monitoring, opening avenues for personalized medicine and optimized therapeutic approaches. Key projects that have contributed to these advancements include IMAGINEHEART and MEDITRAIN, which significantly improved the application of machine learning and medical image processing. IMAGINEHEART focused on developing new methods for cardiovascular image analysis, aiming for precise quantification of key diagnostic parameters such as heart geometry, volume, arterial morphology, and blood flow characteristics. The project resulted in the development of an open-access library for cardiovascular quantifications, enabling the broad application of advanced analytical techniques in both scientific and clinical settings. These methods enhanced the accuracy of cardiac disease diagnostics, improving early detection and intervention. MEDITRAIN introduced an innovative approach to medical education and training for healthcare professionals, combining extended reality (XR) and machine learning to simulate surgical and diagnostic procedures. The development of realistic ultrasound and MRI simulations improved the training of medical personnel, with a special focus on prenatal and neurosurgical diagnostics. These technologies allow for safe and effective practice of complex medical procedures before their application in clinical practice. The continuous research efforts at FERIT are directed toward creating more precise, personalized, and technologically advanced solutions for medical diagnostics. This project builds on previous research and existing research capacities at FERIT, ensuring a high level of feasibility and the potential for significant scientific contributions and knowledge transfer to clinical practice.

By leveraging FERIT's expertise in AI, machine learning, and medical imaging, the proposed project aims to develop new diagnostic tools, improve surgical precision, and enhance patient outcomes, thus positioning itself at the forefront of next-generation medical innovations.

Part e. Risk management strategy

Describe in detail the potential risks related to the project, the impact of risks to the implementation of project activities, and propose ways to mitigate the risks (for each risk separately).

The principal investigator (M. Habijan) will be responsible for conducting both qualitative and quantitative assessments of the project's progress and ensuring the achievement of the defined set of key performance indicators (i.e., results and milestones) by utilizing information provided in internal and periodic reports. The proposed strategy for developing the planned methods is challenging yet feasible. Several potential risks must be considered (as outlined in the table), and these, along with any unforeseen risks, will be managed proactively to prevent critical project delays.

Risk	Probability	Impact	Effect on the project	Mitigation plan
Delays in executing project phases and activities	Low	Low	Odgode.	Complications and delays may occur at any stage of the project. These risks can be mitigated through regular weekly or monthly meetings.
Lack of necessary computational resources	Low	High	Inability to perform key tasks.	Required computational resources should be accounted for in the project budget.
Loss of key human resources	Medium	High	Inability to perform key tasks.	Identify alternative resources in case of unforeseen absences. Assess whether additional human resources can be involved in activities dependent on a single team member. Ensure that complete documentation of all project activities is available at all times.
Competition in research activities	High	High	Inability to publish research findings.	To ensure successful publication, alternative approaches to achieving project objectives should be considered in alignment with the current state of the field.

Part f. Resources

f1. *Describe the resources provided by the Legal entity (if any) for the purpose of conducting the proposed research. List any additional funding sources and submitted or evaluated project proposals that are related to the proposed research.*

The infrastructure currently used for research includes a server with a quad-core processor, 128 GB of RAM, and two Nvidia Titan V GPUs with 5120 CUDA cores each. For data storage, the system is equipped with two 4TB HDDs for medical data storage and two 120GB SSDs for high-speed disk access. The existing infrastructure is primarily used for testing and data storage.

f2. Budget table

CATEGORIES	Year 1	Year 2	Year 3	Year 4	Year 5	TOTAL
Research costs	**17.000,00**	**19.000,00**	**21.000,00**	**21.000,00**	**22.000,00**	**100.000,00**
Equipment costs	**29.999,00**	**20.000,00**	**10.000,00**	**0,00**	**0,00**	**59.999,00**
Training, dissemination, and collaboration costs	**15.000,00**	**15.000,00**	**15.000,00**	**15.000,00**	**15.000,00**	**75.000,00**
Open access publishing costs	**0.00**	**5.000,00**	**5.000,00**	**7.000,00**	**7.000,00**	**24.000,00**
Indirect costs	**9.299,85**	**8.850,00**	**7.650,00**	**6.450,00**	**6.600,00**	**38.849,85**
Total project costs	**71.298,85**	**67.850,00**	**58.650,00**	**49.450,00**	**50.600,00**	**297.848,85**
Senior research assistant's salary	-	-	-	-	-	-

PROJECT BUDGET[2]	71.298,85	65.550,00	54.050,00	49.450,00	58.650,00	298.998,85

Describe the required project budget – *explain all items in the financial plan; describe the required project budget by linking the items in the financial plan to the appropriate objectives and results. The budget should be accurate, clear, justified and realistic.*

The financial plan is based on the actual research needs and ensures that the necessary resources are available to achieve the project's objectives. The budget is structured into several categories, with all expenses directly linked to specific project goals (C1 – C5) and expected outcomes.

The total project cost is €297,848.85, covering field research, professional training for the research team, publication of scientific papers in open-access journals, presentation of project results at international scientific conferences, and the acquisition of new research equipment. ***The project aims to publish four scientific papers in Q1/Q2 Web of Science journals and present intermediate results at seven major scientific conferences, enhancing research visibility and fostering collaboration with potential partners for future projects.*** Emphasis is placed not only on journal publications but also on participation in top-tier international conferences, which are often more relevant than journals in rapidly evolving fields like computing.

Research Costs

The project includes 13 field research trips (up to 14 days for two team members) to leading international universities and research institutes. Since the research team consists of members from three different institutions, ***this initiative enables young researchers to conduct field research that would otherwise be unattainable due to limited institutional funding***. These outgoing mobilities are crucial for achieving high-impact scientific results, providing direct access to cutting-edge knowledge and techniques, which will significantly enhance the project's implementation. Given the interdisciplinary nature of the research and the rapid advancements in deep learning and medical image analysis, continuous training and collaboration with leading global researchers are essential to maintain scientific competitiveness and relevance. Through these mobilities, the team will establish strong research networks with institutions specializing in state-of-the-art histopathological image analysis, medical modality fusion, and explainable AI. ***Regular engagement with international partners will enhance competencies, potentially leading to joint workshops and new EU project applications.*** These activities will increase the visibility of the research team, improve result dissemination, and create a foundation for competitive international collaboration (Cost reference: Items 1.1 – 1.13, linked to objectives C1, C2, C3, C4, and C5 and results R1.4, R2.2, R2.4, R3.2, R3.5, R3.7, R4.1, R4.2, R4.4, R5.2 – R5.7.)

Research Workshop

At the end of the fifth year, the project will host a workshop for medical professionals to present the developed AI-based medical solutions. The project team will introduce medical experts to the innovative systems developed throughout the project and provide training on their application. (Cost reference: Item 1.14, linked to C5 and results R5.3 – R5.8.). Additionally, the project will include eight research group meetings where intermediate results will be presented and assessed, and critical risks identified and addressed. These meetings will take place at IMC Priora, requiring newly acquired equipment for execution. (Cost reference: Items 3.11 – 3.18, linked to C1, C2, C3, C4, and C5, and results R1.4, R2.2, R2.4, R3.2, R3.5, R3.7, R4.1, R4.2, R4.4, R5.2 – R5.7.)

Equipment Costs

The project's primary focus is the development of new algorithms and the application of advanced machine learning methods for medical image processing. ***To achieve this, high-performance workstations with powerful GPUs,*** large RAM capacity, and extensive storage are essential. (Cost reference: Items 2.1 – 2.8, linked to C1, C2, C3, C4, and all expected results.) ***A hyperspectral camera is also required for C3-related activities*** (Results R3.5, R3.7).

Training, Dissemination, and Collaboration Costs

[2] Project budget is a sum of the total project costs and salary of the senior research assistant (salaries for research assistants are not funded under this Call).

The training budget covers team members' visits to leading international universities and research institutes. These training activities are not just beneficial but essential, given the interdisciplinary nature of the project, which merges advanced computational methods with medical applications.
Since the team consists mainly of young researchers, with the planned hiring of an assistant, there is an increased need for continuous professional development in deep learning, medical image processing, biomedical data analysis, and computational diagnostics.
Training will allow the research team to engage with the latest trends and technologies in histological image analysis, multimodal data integration, and explainable AI. Workshops will provide hands-on experience with complex medical datasets, enhancing expertise and the ability to develop innovative solutions. This training will facilitate knowledge exchange with leading researchers and medical professionals, ensuring that developed methods align with real-world clinical applications. (Cost reference: Items 3.1 – 3.10, linked to C1, C2, C3, C4, and results R1.4, R2.2, R2.4, R3.2, R3.5, R3.7, R4.1, R4.2, R4.4, R5.2 – R5.7.)

Conference Presentations at International Scientific Meetings
Based on experience, the average cost per team member for attending a top-tier international conference ranges from €4,000 to €5,000. Over the project duration, the team plans to present research findings at seven international conferences, focusing on C1 and C2-related objectives (Results R1.2, R1.3, R1.4, R1.6). Additionally, two conference papers will be presented for C3-related objectives (Results R3.2, R3.7), with two more for C4 objectives (Results R4.1, R4.2). (Cost reference: Item 5.1.).

Journal Publications
Given the team size and project duration, four scientific papers will be published in Q1/Q2 journals, with an allocated budget of €24,000. (Cost reference: Items 4.1 – 4.4, linked to C1, C2, C3, C4, and C5, and results R2.2, R3.5, R4.4, R5.8.). Leading open-access journals in medical image processing have significantly higher publication fees than closed-access journals, often ranging from €5,000 to €10,000 per article. Thus, €24,000 is allocated for four journal publications over five years.
This budget ensures that the project is well-supported in terms of research mobility, computing resources, professional training, dissemination, and collaboration, maximizing its impact and long-term sustainability.

Part g. Ethical issues

In the table below, indicate whether the proposed research raises ethical issues. If any ethical issue is indicated, the Ethics Issues Checklist should be submitted at the time of application, whereas the approval of the relevant ethics committee should be submitted during evaluation of the project proposal, or at the latest, before singing the grant contract. The Ethics Issues Checklist is available on the HRZZ website.

Please indicate if the proposed research includes the following:	
X	Human embryonic stem cells and human embryos, human cells or tissues
X	Humans
X	Data collection and processing (privacy)
X	Animals
X	Environment, health and safety
X	Artificial intelligence
X	Other ethics issues and possible misuse of research results

Part B - Principal Investigator (PI)

Part a. Track record of the principal investigator (PI) in the last 5 years (maximum 2 pages).

1	*List the five most important peer-reviewed publications in international journals in which the PI is the first or corresponding author, or other works that impacted the PI's research area, such as articles in journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art works and exhibitons, strategies, contributions to policy making etc).*
	1. Benčević, Marin; Habijan, Marija; Galić, Irena; Babin, Danilo; Pižurica, Aleksandra, „Understanding skin color bias in deep learning-based skin lesion segmentation", Computer methods and programs in biomedicine, 245 (2024), 0; 108044, 10, doi: 10.1016/j.cmpb.2024.108044 ***Q1, IF: 6.1, Google scholar citati: 13***
	2. Hartmann Tolić, I.; Habijan, M.; Galić, I.; Nyarko, E.K. Advancements in Computer-Aided Diagnosis of Celiac Disease: A Systematic Review. Biomimetics 2024, 9, 493, https://doi.org/10.3390/biomimetics9080493 ***Q1, IF: 3.4, Google scholar citati: 2***
	3. Benčević, Marin; Galić, Irena; Habijan, Marija; Babin, Danilo. „Training on Polar Image Transformations Improves Biomedical Image Segmentation", IEEE Access, 9 (2021), 133365-133375, doi:10.1109/access.2021.3116265 , ***Q1, IF: 3.557, Google scholar citati: 62***
	4. Habijan, Marija; Galić, Irena; Leventić, Hrvoje; Romić, Krešimir „Whole Heart Segmentation Using 3D FM-Pre-ResNet Encoder–Decoder Based Architecture with Variational Autoencoder Regularization", Applied Sciences, 11 (2021), 9; 3912, 21 doi:10.3390/app11093912 ***Q2, IF: 2.7, Google scholar citati: 18***
	5. Habijan, Marija; Galić, Irena; Romić, Krešimir; Leventić, Hrvoje, „AB-ResUNet+: Improving Multiple Cardiovascular Structure Segmentation from Computed Tomography Angiography Images", Applied Sciences, 12 (2022), 6, doi: 10.3390/app12063024, ***Q2, IF: 2.7, Google scholar citati: 14***
2	*Previous participation in HRZZ projects (role in the project, key publications and other important results from the project).*
	HRZZ1. Research associate on the project "Medical Education and Planning with Extended Reality" (MEDITRAIN), funded by the Croatian Science Foundation (HRZZ), Research Project (December 16, 2024 – December 15, 2027). **HRZZ2.** PhD student (since May 2, 2018) on the project "Methods for the Interpretation of Medical Images for Detailed Analysis of Heart Health" (IMAGINEHEART), funded by the Croatian Science Foundation (HRZZ), Establishing Research Project (March 1, 2018 – February 28, 2023). The project resulted in **9** scientific papers published in Q1/Q2 journals, **14** papers presented at international scientific conferences, **3** joint PhDs (Hrvoje Leventić, Marija Habijan, Marin Benčević), and **17** software codes. The final project evaluation was A - Excellent Progress (the project fully achieved its goals within the given timeframe or exceeded expectations). The most important publications from the IMAGINEHEART project are listed in the table above.
3	*Other research projects (role in the project, funding source, allocated funds, key publications and other important results from the project, with links to the websites of the funded projects).*
	"Modern AI Architectures for Medical Applications - Fusion of Computer Vision and Large Language Models", Role: Research Associate, National Recovery and Resilience Plan, 2025. "Digital Health and Patient Safety Platform (DHPSP)", Ludwig Boltzmann Institute for Digital Health and Patient Safety (LBIDHPS), Medical University of Vienna, Research Associate, 2021 – 2024. "Analysis and Reduction of Bias in Medical Image Segmentation", 2023 – 2024, Research Associate, J.J.StrossmayerUniversityofOsijek. "Analysis of Abdominal Aortic Aneurysm Using Computational Fluid Dynamics for Predicting Rupture Risk", 2022 – 2023, Research Associate, J.J. Strossmayer University of Osijek. "Detection of Risk for Post-Operative Complications After LAA Occlusion", 2019 – 2020, Research Associate, J.J. Strossmayer University of Osijek.

4	*Other major academic accomplishments: invited lectures at international conferences and/or summer schools, organisation of scientific meetings, international awards, membership in editorial boards of journals, peer review of articles for international journals, evaluation of research projects, major scientific collaborations, collaborations with industry.*
	The organization of international scientific conferences includes participation as a member of the organizing committee for the International Conference on Systems, Signals and Image Processing 2019 (IWSSIP 2019) and the International Conference on Smart Systems and Technologies in 2018 and 2020 (SST2018 and SST2020). Editorial board memberships include serving as a guest editor for MDPI Electronics in the special issue "Revolutionizing Medical Image Analysis with Deep Learning, 2nd Edition," along with Irena Galić, Daniel Vasić, and Mirela Kundid Vasić. Additionally, guest editing was conducted for the first edition of the special issue "Revolutionizing Medical Image Analysis with Deep Learning," in collaboration with Irena Galić, Antonio Lanata, and Luca Mesin. Peer review activities for international scientific journals include reviewing for Applied Artificial Intelligence (1), Applied Sciences (21), Axioms (1), Computers (1), Diagnostics (3), Electronics (8), IEEE Access (11), Journal of Imaging (7), Journal of Personalized Medicine (2), Mathematics (3), Medicina (2), PloS One (2), Review of Scientific Instruments Online (2), Sensors (9), and Technology in Cancer Research and Treatment (11). Key scientific collaborations involve work with ***Prof. Dr. Ir. Aleksandra Pižurica*** from Ghent University, Faculty of Engineering and Architecture, GAIM; ***Dr. Ir. Danilo Babin*** from Ghent University, Faculty of Engineering and Architecture, imec-TELIN; Assist. ***Prof. Dr. Josip Samardžić***, MD, from ***General Hospital Dr. Josip Benčević in Slavonski Brod***; ***Tomislav Kalem***, MD, from IMC Priora; ***Dr. Srđan Lazendić*** from the Faculty of Engineering and Architecture, Clifford Research Group; and ***Petar Nakić***, MSc. Comp. Eng., from the Technical Department at the University of Slavonski Brod.
5	*Highlight your previous scientific contribution in the research field and topic and in mentoring early career researchers (provide the number of successfully defended doctoral dissertations and joint publications).*
	The applicant has participated in the development of new machine learning methods for processing and analyzing various cardiovascular structures from X-ray, CT, and MR images. Throughout her scientific research career, particularly during her doctoral studies and the implementation of the Croatian Science Foundation (HRZZ) project "Methods for the Interpretation of Medical Images for Detailed Analysis of Heart Health," the applicant has made a significant contribution to the research field in which this project proposal is being conducted. During her dual doctoral studies and training at the Faculty of Engineering and Architecture, she spent six months at Ghent University. Currently, she is not a PhD supervisor but has successfully co-supervised two PhD students, Marin Benčević (2024) and Filip Bajić (2024), one of whom, Dr. Marin Benčević, defended a dual PhD dissertation in collaboration with Ghent University in 2024. The applicant closely collaborated with PhD students, resulting in a total of five papers published in Web of Science-indexed scientific journals and six papers presented at international scientific conferences. These achievements highlight her success in mentoring young researchers. The complete list of publications is available in the CRORIS database and on her Google Scholar profile. In addition to these accomplishments, the applicant has received three awards for her scientific research work: the "Vera Johanides" Young Scientist Award from the Croatian Academy of Technical Sciences (2024), the "Outstanding Young Scientist" award from FERIT Osijek (2023), and a recognition for excellence in postgraduate university studies at FERIT Osijek (2022). She has been a member of the Elmar Society since 2022 and the Medical Image Computing and Computer-Assisted Intervention (MICCAI) Society since 2024.

1b) Curriculum vitae of the Principal Investigator (PI) (maximum 2 pages, according to the following template).

1	***PERSONAL INFORMATION***	
	First name and last name:	Marija Habijan
	Researcher's national ID[3]:	370130
	Website (ORCID and/or CROSBI):	*https://orcid.org/0000-0002-3754-498X*
2	***EDUCATION***	
	2022 PhD	*PhD in Computer Science (Joint PhD)*
	Institution:	*University of J.J. Strossmayer Osijek, Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek & Ghent University, Faculty of Engineering and Architecture, Belgium*
	2015 mag.ing.comp.	*MSc in Computer Engineering*
	Institution:	*Faculty of Electrical Engineering, Osijek, Croatia*
3	***POSTDOCTORAL TRAINING***	
	2024 Institution:	*Ghent University, Faculty of Engineering and Architecture*
	Funding source:	*Erasmus+ (30 days)*
4	***EMPLOYMENT***	
	2022 – present	*Senior Assistant*
	Institution:	*University of J.J. Strossmayer Osijek, Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
	2024 – present	*Research Associate*
	Institution:	*International Medical Center Priora*
5	***Previous employment***	
	2018 – 2022 - position	*Assistant*
	Institution:	*University of J.J. Strossmayer Osijek, Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
	2017 – 2018 - position	*DevOps Engineer*
	Institution:	*Styria IT Solutions, Zagreb*
	2015 – 2017 - position	*Application Engineer*
	Institution:	*Croatian Telecom, Zagreb*
6	***SCHOLARSHIPS AND AWARDS***	
	2024 - Croatian Academy of Technical Sciences	*"Vera Johanides" Young Scientist Award*
	2023 - FERIT Osijek	*"Outstanding Young Scientist"*
	2022 - FERIT Osijek	*Recognition for Excellence in Postgraduate Studies*
7	***MENTORING AND CO-MENTORING***	
	2018 – 2024 Number of master theses defended / number of dissertations defended / number of postdocs	*Co-supervised 2 defended PhD dissertations (Marin Benčević, Filip Bajić)* *Co-supervised 8 defended master's and bachelor's theses*
	Institution	*University of J.J. Strossmayer Osijek, Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
8	***TEACHING ACTIVITIES***	
	2018 – present, computer science	*Image Processing and Computer Vision (Lab), Quantum Computing (1/3 Lecture, Seminar, Lab), 3D Computer*

[3] If applicable

		Graphics (Lab), Digital Signal Processing (Seminar, Lab), Signals and Systems (Seminar, Lab)
	Institution:	*Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
	2018 – present, computer science	*Information Theory (Seminar, Lab), Computer Vision*
	Institution:	*Faculty of Natural Sciences, Mathematics, and Educational Sciences, Mostar, Bosnia and Herzegovina*
9	***COMMITMENTS IN THE INSTITUTION (if applicable)***	
	201? – Council member, name of institution/state	***Not applicable***
10	***MEMBERSHIPS (e.g., scientific committees or associations, evaluation committees, etc.; indicate year and institution) (if applicable)***	
	Member of Elmar Society (since 2022) *Member of the Medical Image Computing and Computer-Assisted Intervention (MICCAI) Society (since 2024)*	
11	***CAREER BREAKS (if applicable)***	
	Indicate duration (from-to in dates) and reason	

Part C - Research group and project management

Part a. Research group:

*List in the table all members of the research group who will participate in the proposed research; indicate the Principal Investigator (**PI**) and ALL team members (senior research assistants (**PD**), research assistants (**D**) and students (**S**)*. Explain the role of each member of the research group in the project.*

Members of the research group[4]

Add lines as needed

First and last name	Title	Organisation (country)	PhD defence year**	Project role (PI, D, PD, S)	Time commitment (%)
Marija Habijan	dr. sc.	Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek	2022.	V	55%
Marija Habijan (PI) will dedicate 55% of her working time at FERIT to the project. Her primary responsibility is to coordinate all project activities and ensure the successful implementation of all segments. She will be directly involved in the following tasks: - Hiring and supervising an assistant, - Organizing the procurement of computers and related equipment, - Preparing periodic and final project reports, - Collecting clinical data in collaboration with IMC Priora Hospital (C1, C2, C3), - Developing methods for the automatic segmentation of neurovascular structures and tumor tissue (C3), - Implementing attention-based architectures to improve model interpretability (C5), - Participating in international scientific conferences (C2, C4), - Developing deformable registration models for hyperspectral analysis (C3), - Evaluating and validating AI models in clinical practice (C5), - Writing scientific papers. Additionally, she will ensure that all research activities and resulting publications comply with ethical standards for scientific research involving humans and patients in Croatia. In the final year of the project, she will conduct a comprehensive analysis, identify risks, and define new research directions based on completed activities. Furthermore, she will actively seek potential funding sources for new project proposals.					
Hrvoje Leventić	doc. dr. sc	Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek	2019.	S	15%
Hrvoje Leventić will dedicate 15% of his working time at FERIT to the project. He will be directly involved in the following activities: • Developing methods for the segmentation of suspicious prostate lesions using deep neural networks (C2), • Multimodal registration of MRI and ultrasound using deformable neural networks (C2), • Implementing attention-based architectures for model interpretability (C5), • Contributing to the development of ROBOMED-Fusion and ROBOMED-VideoAI software frameworks (C2, C4), • Validating AI models in clinical practice (C5).					
Marin Benčević	dr. sc.	Faculty of Electrical Engineering, Computer Science, and	2024.	VA	25%

[5] Principal investigators who submit bilateral or trialteral project proposals should list members of the research partner group in a separate table (by copying the provided table template).

		Information Technology Osijek			
Marin Benčević will dedicate 25% of his working time at FERIT to the project. He will be directly involved in the following activities: • Developing methods for the segmentation of suspicious prostate lesions (C2), • Multimodal registration of MRI and ultrasound (C2), • Developing methods for 3D prostate model generation for tumor lesion visualization (C2), • Implementing attention-based architectures to enhance model interpretability (C5), • Participating in international conferences (C2), • Collecting data for hyperspectral analysis and validating models (C3).					
Srđan Lazendić	**dr. sc.**	**Faculty of Engineering and Architecture, Ghent University**	**2023.**	**VA**	**40%**
Srđan Lazendić will dedicate 35% of his working time at Ghent University to the project. He will be directly involved in the following activities: • Developing methods for segmentation of suspicious prostate lesions using deep neural networks (C2), • Implementing methods for multimodal registration of MRI and ultrasound images using deformable neural networks (C2), • Contributing to a scientific paper on multimodal MRI and ultrasound registration for fusion biopsy of the prostate (C2), • Participating in international scientific conferences related to fusion biopsy of the prostate (C2), • Developing methods for 3D prostate model generation for tumor lesion visualization (C2), • Contributing to the development of the ROBOMED-Fusion framework for multimodal registration of MRI and ultrasound data in fusion biopsy of the prostate (C2).					
Josip Samardžić	**dr.sc. / dr. med.**	**General Hospital Dr. Josip Benčević, Slavonski Brod**	**2015.**	**Liječnik /suradnik**	**15%**
Josip Samardžić will dedicate 15% of his working time at General Hospital Dr. Josip Benčević in Slavonski Brod to the project. He will be directly involved in the following activities: • Collecting clinical data in collaboration with IMC Priora Hospital (C1, C2, C3), • Developing methods for intraoperative visualization of segmented tissues (C3), • Participating in the evaluation and validation of AI models (C5), • Developing methods for tracking intraoperative instrument movements (C4).					
Danijel Marinčić	**dr. med. urolog**	**General Hospital Dr. Josip Benčević, Slavonski Brod, IMC Priora, Čepin**	**doctoral student**	**Liječnik /suradnik**	**30%**
Danijel Marinčić will dedicate 30% of his working time at General Hospital Dr. Josip Benčević in Slavonski Brod and IMC Priora to the project. He will be directly involved in the following activities: • Collecting clinical data in collaboration with IMC Priora Hospital (C1, C3), • Contributing to the development of methods for intraoperative visualization of segmented tissues (C3), • Developing methods for the analysis of intraoperative video recordings of robotic surgery (C4), • Evaluating AI models in clinical practice (C5).					
Branimir Bošnjak	**dr. med. urolog**	**General Hospital Dr. Josip Benčević, Slavonski Brod, IMC Priora, Čepin**	-	**Liječnik /suradnik**	**30%**
Branimir Bošnjak will dedicate 30% of his working time at General Hospital Dr. Josip Benčević in Slavonski Brod to the project. He will be directly involved in the following activities: • Developing methods for processing and analyzing histopathological images (C1), • Contributing to the technical report on generative models for virtual staining (C1),					

• Developing methods for multimodal registration of MRI and ultrasound (C2), • Participating in the development of the ROBOMED-HistoAI and ROBOMED-Fusion software frameworks (C1, C2), • Validating AI models in clinical practice (C5).					
Petar Nakić	**mag. ing. comp.**	**University of Slavonski Brod, Technical department**	**doctoral student**	**I**	**35%**
Petar Nakić will dedicate 35% of his working time at the University of Slavonski Brod to the project. He will be directly involved in the following activities: • Developing methods for processing histopathological images and virtual staining (C1), • Integrating multimodal data into the artificial intelligence system (C1), • Contributing to the development of the ROBOMED-HistoAI software framework (C1).					
Maja Čuletić Čondrić		**University of Slavonski Brod, Tehnički odjel**	**doctoral student**	**I**	
Maja Ćuletić Čondrić will dedicate 35% of her working time at the University of Slavonski Brod to the project. She will be directly involved in the following activities: • Developing methods for processing and analyzing histopathological images (C1), • Multimodal registration of MRI and ultrasound (C2), • Contributing to the development of methods for intraoperative video analysis (C4), • Collecting clinical data in collaboration with IMC Priora Hospital (C1, C2, C3).					
Nives Grčić	**bacc. ing. comp.**	**Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek**	**-**	**S**	**20%**
Nives Grčić will dedicate 20% of her working time to the project. She will be directly involved in the following activities: • Contributing to the technical report on state-of-the-art machine learning methods for histopathological image analysis (C1), • Developing methods for processing and analyzing histopathological images (C1), • Modeling relationships between tumor regions and the microenvironment in histopathological images (C1), • Participating in international scientific conferences related to histopathological image processing methods (C1), • Developing methods for segmenting suspicious prostate lesions using deep neural networks (C2), • Contributing to the development of multimodal MRI and ultrasound registration methods using deformable neural networks (C2), • Contributing to technical reports related to clinical data collection in collaboration with IMC Priora Hospital (C1).					
PhD student	**-**	**-**	**-**	**A**	**100%**
The PhD student employed on the project will dedicate 100% of their working time at FERIT. They will be directly involved in the following activities: • Participating in the collection and processing of clinical data (C3, C4), • Developing of all methods for intraoperative visualization of segmented tissues (C3), • Contributing to the development of all methods for tracking intraoperative instrument movements (C4), • Evaluating and validating AI models in clinical practice (C5).					

*For team members who are not employed at the time of application, but are planned, indicate the project role (D or PD) in the "First and Last Name" column, add time committment in percentage, and describe the tasks in the project.
** If applicable.

Part b. Project management plan

Describe in detail competences and research activities of research group members, managing plan for the research group to achieve research objectives and results for each project period (in relation to the plan for publishing and

dissemination, the plan for employing and training early career researchers). Describe the plan for procurement of new equipment and the use of this equipment to achieve the objectives.

The research team for this project ***brings together interdisciplinary researchers*** with expertise in computer science, medical diagnostics, mathematical modeling, and biomedical engineering. Their competencies have been developed through previous scientific projects, collaborations with medical professionals, and research conducted during the doctoral studies of team members. The research will primarily be conducted at Josip Juraj Strossmayer University of Osijek, Faculty of Electrical Engineering, Computer Science, and Information Technology (FERIT). FERIT will allocate part of the working time of its employees involved in the project.

The project leader, Marija Habijan, will coordinate research activities, oversee the preparation of scientific publications, manage the dissemination of results, and mentor junior researchers engaged in the project. She will also be actively involved in developing key methods required to achieve the project goals and will contribute to the implementation of methods necessary for C1, C2, C3, C4, and C5.

All team members will participate in research, development, validation of methods, writing scientific papers, and preparing documentation related to implemented algorithms and analyses. The project also involves an international collaboration with researchers from Ghent University, Faculty of Engineering and Architecture (Dr. Srđan Lazendić), with whom FERIT Osijek has collaborated for many years.

The project will include cooperation with IMC Priora (Danijel Marinčić, M.D.), which possesses medical devices such as Histolog Scanner, Mona Lisa, and Da Vinci, enabling clinical data collection and execution of research activities outlined in the project. The project will also involve medical specialists conducting clinical research at General Hospital Dr. Josip Benčević in Slavonski Brod (Assoc. Prof. Josip Samardžić, Ph.D., Branimir Bošnjak, M.D.), ensuring validation of the developed methods in real medical scenarios.
A part of the research, focusing on histopathological image processing, analysis, and model optimization, will be carried out at the University of Slavonski Brod (Petar Nakić, Maja Čuletić Čondrić).

Competences and scientific research activities of members of the research group

The research team consists of an interdisciplinary group of experts in computer science, medical image processing and analysis using machine learning methods, mathematical modeling, and biomedical engineering. The key members of the research group and their competencies include:

Assoc. Prof. Hrvoje Leventić, Ph.D. – Associate Professor at FERIT, possesses exceptional expertise in computer engineering and medical image processing. His research experience includes working on advanced methods for segmentation and analysis of medical images, particularly in the cardiovascular system. Among his most important publications is a study on the segmentation of the left atrium from 3D CCTA images for occluder placement. During his scientific training at Ghent University, he collaborated with top researchers, gaining expertise in developing advanced methods for cardiovascular structure segmentation, culminating in his joint Ph.D. defense. In addition to his outstanding scientific contributions, Dr. Leventić received the "Vera Johanides" award for young scientists in 2021 from the Croatian Academy of Technical Sciences. His work in this project contributes to the application of advanced computer vision and machine learning methods for medical image processing and analysis.

Marin Benčević, Ph.D. – Senior Assistant at FERIT, specializes in medical image analysis and machine learning methods. His research is dedicated to advanced studies in medical image processing and machine learning applications. His scientific work includes numerous publications focusing on various aspects of medical image analysis. A key contribution of his work is medical image segmentation, where he developed innovative methods for more accurate and efficient extraction of relevant structures from imaging data. He paid special attention to reducing skin color bias in lesion segmentation, which is crucial for accurate diagnosis and treatment of dermatological conditions across different populations. His research also includes improving biomedical image segmentation using polar transformations, enabling better understanding and analysis of anatomical structures. His work on understanding skin color bias in deep learning-based lesion segmentation analyzed how different skin tones affect model performance and proposed methods to mitigate such biases.

Srđan Lazendić, Ph.D. – Researcher at Clifford Research Group, Ghent University. He graduated in pure mathematics with a teaching specialization from the University of Novi Sad, Serbia (2016), and earned his

Ph.D. in mathematical engineering from Ghent University (2023). His research interests focus on applying Clifford algebra methods for efficient analysis of high-dimensional data and image processing. He also works on nonlinear optimization methods and explainable deep learning models based on sparse representations. Recently, he has begun researching functional-analytical methods for wave propagation models in viscoelastic media.

Assoc. Prof. Josip Samardžić, M.D., Ph.D. – Director of General Hospital "Dr. Josip Benčević" in Slavonski Brod, a distinguished surgeon and scientist with extensive experience in clinical research, surgery, and post-operative patient outcome analysis. His years of experience managing hospital institutions and active involvement in international research projects facilitate the integration of computational image analysis with clinical practice. His expertise includes digestive surgery, endoscopic surgery, and emergency medicine, and he has significantly contributed to research on post-operative complications and surgical process optimization. As an associate professor at the Faculty of Dental Medicine and Health Osijek, he actively participates in educating future doctors and researchers. Through membership in professional and scientific associations such as the European Association of Endoscopic Surgery (EAES) and the American College of Surgeons, he contributes to medical practice advancement and international knowledge exchange. His extensive experience in analyzing and interpreting medical data will ensure that computational models are tailored to clinical practice needs.

Danijel Marinčić, M.D. – Doctoral student in biomedicine at the Ruđer Bošković Institute and urology specialist with extensive experience in medical diagnostics and advanced surgical techniques. His work focuses on robotic surgery, minimally invasive procedures, and 3D modeling in surgical planning, improving the precision and efficiency of surgical procedures. He is the head of the surgical block and central sterilization at General Hospital "Dr. Josip Benčević" in Slavonski Brod and a urology specialist at IMC Priora. His career includes participation in numerous national and international training programs, such as Intuitive DaVinci Multiport technology and IRE master class therapy for prostate cancer, equipping him with expertise in advanced surgical technologies. As a member of the European Association of Urology and the Croatian Urological Society, he actively engages in scientific research and professional development. His role in this project is crucial for ensuring the clinical relevance of developed technologies, validating methods, and integrating innovative solutions into surgical practice.

Branimir Bošnjak, M.D. – Specialist in medical diagnostics and urology with extensive experience in minimally invasive surgical techniques, particularly in robotic surgery and innovative surgical planning methods. His work focuses on enhancing surgical precision, reducing post-operative complications, and improving patient outcomes through advanced technologies. Throughout his career, he has gained extensive clinical experience in urology, oncological surgery, and reconstructive procedures. His professional engagement includes continuous education, participating in international training sessions and workshops on robot-assisted surgery, laparoscopy, and personalized surgical approaches.

Petar Nakić, M.Sc. – Doctoral student in computer science at the University of Split and a lecturer at the Technical Department, University of Slavonski Brod. He holds a master's degree in computer science from the University of Osijek (2015). His research interests include data security and electronic data processing in biomedicine. As a lecturer, he actively contributes to teaching undergraduate and graduate students, transferring knowledge and expertise in computer science and information technology.

Maja Čuletić Čondrić, Ph.D. – Mathematician and senior lecturer with extensive experience in statistical analysis and data modeling, crucial for validating results and analyzing data in this project. She currently works as a senior lecturer at the Polytechnic of Slavonski Brod, teaching mathematics and physics and participating in research projects related to mathematical methods for data analysis. Her expertise includes numerical mathematics, geometric and fractal structure analysis, and applying statistical models in natural and technical sciences. She is the author of numerous scientific papers published in international journals and is interested in mathematical model optimization, fractal algorithm applications, and quantitative data assessment methods.

The research team's multidisciplinary expertise ensures a comprehensive approach to developing advanced artificial intelligence methods for medical diagnostics and robotic surgery. Their combined knowledge in

machine learning, image processing, clinical practice, mathematical modeling, and biomedical engineering provides a strong foundation for achieving the project's objectives.

Scientific research activities of collaborators and research group management plan

Marija Habijan leads the project and coordinates research activities. She actively participates in the collection of clinical data in collaboration with IMC Priora hospital (C1, C2, C3) and in the development of methods for the automatic segmentation of neurovascular structures and tumor tissue (C3). She also works on the implementation of attention-based architectures to improve model interpretability (C5). She participates in international scientific conferences related to fusion biopsy of the prostate and the automatic analysis of intraoperative video recordings (C2, C4) and leads the development of deformable registration models for hyperspectral analysis (C3). In the final phases of the project, she will be involved in the evaluation and validation of developed AI models in clinical practice (C5).

Hrvoje Leventić participates in the development of methods for segmenting suspicious prostate lesions using deep neural networks (C2) and in the implementation of methods for multimodal registration of MRI and ultrasound images using deformable neural networks (C2). He actively contributes to the implementation of attention-based architectures for model interpretability (C5) and the development of software frameworks ROBOMED-Fusion and ROBOMED-VideoAI (C2, C4). Additionally, he participates in the validation of developed AI models in clinical practice (C5) and presents research results at international conferences.

Marin Benčević is involved in the development of methods for segmenting suspicious prostate lesions (C2) and multimodal registration of MRI and ultrasound (C2). He contributes to the creation of 3D prostate models for visualizing tumor lesions (C2) and the implementation of attention-based architectures for model interpretability (C5). His role also includes the development of methods for hyperspectral analysis of intraoperative data (C3). He actively participates in scientific publications and conferences related to precise medical diagnostics.

Srđan Lazendić works on the development of methods for segmenting suspicious prostate lesions using deep neural networks (C2) and the implementation of methods for multimodal registration of MRI and ultrasound using deformable neural networks (C2). He participates in research related to optimizing the fusion biopsy procedure of the prostate and in the creation of 3D prostate models for visualizing tumor lesions. In the final phases of the project, he contributes to the development of the ROBOMED-Fusion software framework for MRI and ultrasound image registration (C2).

Josip Samardžić is responsible for collecting clinical data in collaboration with IMC Priora hospital (C1, C2, C3) and developing methods for intraoperative visualization of segmented tissues (C3). He participates in the evaluation and validation of AI models and in the development of methods for tracking intraoperative movements of surgical instruments (C4). Additionally, he presents research results at international conferences and contributes to the final evaluation of the clinical application of developed AI models (C5).

Danijel Marinčić participates in the collection of clinical data in collaboration with IMC Priora hospital (C1, C3) and the development of methods for intraoperative visualization of segmented tissues (C3). His role also includes implementing methods for analyzing intraoperative video recordings of robotic surgery (C4). He actively contributes to the validation of developed AI models and participates in the dissemination of results through scientific publications and conferences.

Branimir Bošnjak participates in the development of methods for processing and analyzing histopathological images (C1) and in the implementation of generative models for virtual staining of medical images. He is involved in work on multimodal registration of MRI and ultrasound images (C2) and the development of the ROBOMED-HistoAI and ROBOMED-Fusion software frameworks (C1, C2). Additionally, he participates in the validation of developed AI models and their application in clinical practice (C5).

Petar Nakić is involved in the development of methods for processing histopathological images and applying generative models for virtual staining (C1). He participates in integrating multimodal data into the artificial intelligence system and developing the ROBOMED-HistoAI software framework for histopathological image analysis. He also contributes to technical reports and scientific publications related to improving cancer diagnostics.

Maja Ćuletić Čondrić participates in developing methods for processing and analyzing histopathological images (C1), including modeling relationships between tumor regions and the microenvironment. She contributes to methods for multimodal registration of MRI and ultrasound (C2) and the implementation of intelligent systems for analyzing intraoperative video recordings (C4). She actively participates in the collection of clinical data and the development of AI models.

Nives Grčić contributes to technical reports on state-of-the-art machine learning methods for histopathological image analysis (C1) and the development of methods for processing and analyzing histopathological data, including modeling relationships between tumor regions and the microenvironment. She is also involved in the development of methods for segmenting suspicious prostate lesions using deep neural networks (C2) and multimodal registration of MRI and ultrasound (C2). She participates in international scientific conferences and the dissemination of research results.

The PhD student participates in the collection and analysis of clinical data (C3, C4), the development of methods for intraoperative visualization of segmented tissues (C3), and tracking intraoperative instrument movements (C4). Additionally, they contribute to the evaluation and validation of developed AI models in clinical practice (C5) and are actively involved in developing new methods and scientific publications. A proposal will be submitted for the HRZZ project "Career Development Project for Young Researchers – Training of New PhDs" and hiring a PhD student whose dissertation topic would be closely related to the project. Depending on the announcements of HRZZ project calls, another application may be submitted in the third year of the project if it aligns with the project call's time constraints and duration.

Results publication and dissemination plan

The ROBOMEDON project focuses on the development of advanced machine learning methods for medical diagnostics and robotic surgery. To achieve the defined objectives (C1, C2, C3, C4, and C5), a series of results are planned to be achieved over a five-year period, ensuring their dissemination through scientific papers, conferences, databases, and industrial collaboration.

In the first year, the focus will be on the preparation of technical reports on existing machine learning methods for histopathological image analysis (C1), MRI-ultrasound fusion (C2), and intraoperative hyperspectral analysis (C3). Additionally, initial models for the segmentation of histological structures and tumor lesions, as well as the first methodologies for MRI and ultrasound data fusion, will be developed. To disseminate the results, the team will participate in international conferences to present initial findings on medical image analysis methods. Furthermore, a scientific paper on the segmentation of histopathological images will be submitted for peer review in a relevant journal.

In the second year, a scientific paper on an algorithm for multimodal registration of MRI and ultrasound data (C2) will be published in a high-impact journal. Additionally, researchers will participate in international conferences presenting results related to hyperspectral analysis of intraoperative data (C3) and automated surgical video analysis (C4). Dissemination through these channels will facilitate connections with global researchers and potential collaborations for further method development.

In the third year, the focus will be on the clinical validation of the developed models and software frameworks ROBOMED-HistoAI, ROBOMED-Fusion, and ROBOMED-VideoAI. The results of these validations will be published in leading medical and computational journals (Q1/Q2), and workshops will be organized for medical professionals to enhance the practical application of the developed methods. At the same time, a scientific paper on hyperspectral analysis methods for intraoperative detection of neurovascular structures (C3) will be submitted for peer review. International conferences will feature presentations on segmentation and data registration methods, while open access to the developed models will enable broader application.

The fourth year will be crucial for result dissemination through the publication of papers on model interpretability in medical practice (C5) and the integration of artificial intelligence into clinical systems. Special attention will be given to organizing educational workshops for medical personnel to facilitate the implementation of developed AI models. Additionally, a scientific paper on intraoperative video analysis methods for robotic surgery will be published in prestigious medical journals. The project team plans to participate in international conferences to present results in the field of surgical video processing and real-time AI models.

In the fifth and final year, the focus will be on the final evaluation and publication of results through scientific papers and final technical reports. The software frameworks ROBOMED-HistoAI, ROBOMED-Fusion, ROBOMED-HyperSpectral, ROBOMED-VideoAI, and ROBOMED-XAI will be finalized and presented to the research and industrial communities. A scientific paper on the integration of AI models into clinical information systems (C5) will be published, and the developed methods will be included in publicly available databases. International conferences will showcase the evaluation results of the models and their application in clinical practice. The final phase of the project will focus on collaboration with industrial partners to explore potential commercialization of the research findings.

Throughout the project, open access to data, code, and models will be ensured, enabling broad application of the research results. Dissemination through scientific papers, conferences, and collaborations with medical institutions will secure the long-term scientific and societal impact of the ROBOMEDON project.

Employment and training plan for newly employed doctoral students

A project application for the Croatian Science Foundation (HRZZ) program "Career Development of Young Researchers – Training of New Doctoral Students" is planned, with the aim of employing a doctoral candidate whose dissertation topic will be closely related to the project's scope. Depending on the calls issued under the HRZZ program, an additional application may be submitted in the third year of the project, provided that the timing aligns with the project call constraints and project duration.

The newly hired doctoral candidate D will be engaged in the project full-time. They will enroll in a dual doctoral program at their home institution (FERIT) and Ghent University, with a dissertation topic directly aligned with the project's objectives. The candidate will be involved in the development and optimization of models for fusion biopsy of the prostate, enhancing the integration of MRI and ultrasound data into diagnostic procedures. They will contribute to the design, development, and implementation of machine learning methods for histopathological image analysis, improving the accuracy and efficiency of tumor tissue diagnostics.

Additionally, the doctoral candidate will work on developing methods for segmenting neurovascular structures during robotic surgical procedures, improving the safety and precision of operations. In parallel, they will contribute to the development and implementation of models for intraoperative video analysis in robotic surgery using deep learning and computer vision. The candidate will develop a system for automatic recognition of surgical instruments and their interactions with tissue, enhancing the understanding of surgical procedures and providing better intraoperative support to surgeons.

Furthermore, the doctoral candidate will be involved in the development of a modular framework for explainability in AI models for medical applications, enabling clinicians to better understand AI-driven decisions. They will also develop software for analyzing hyperspectral intraoperative data, enhancing real-time detection of tumor tissues and neurovascular structures. The candidate will contribute to the creation of a comprehensive database of histopathological and intraoperative images, providing valuable resources for future research and AI model validation. Their work in applying synthetic datasets and advanced machine learning techniques will advance medical analytics and improve diagnostic and surgical processes.
Finally, the doctoral candidate will actively participate in writing scientific papers, presenting their research findings to the academic community, thereby increasing the project's visibility in the scientific domain. This management and training plan will ensure a comprehensive professional development path for the doctoral candidate, preparing them for future challenges in medical image processing, analysis, and synthesis research.

Equipment procurement plan and use for the purpose of achieving goals

As part of the ROBOMEDON project, the procurement of specialized computing and medical equipment is planned to enable research in the field of machine learning for medical diagnostics and robotic surgery. The equipment will support the analysis of histopathological images, multimodal integration of MRI and ultrasound data, intraoperative hyperspectral analysis, and intelligent analysis of intraoperative video recordings.

One of the key procurements includes workstations, which will allow training of complex neural networks on large medical datasets (C1, C2, C3). This equipment will be fundamental for processing histopathological images, segmenting tumor lesions, and developing methods for fusion biopsy of the prostate.
High-performance workstations will also be acquired for the analysis of multimodal medical data (C2), including the registration of MRI and ultrasound images of the prostate using deformable neural networks. These workstations will enable the optimization of algorithms and faster validation of models on large datasets.

For intraoperative analysis, the procurement of a hyperspectral camera is planned, which will enable advanced recognition of neurovascular structures during robotic surgeries (C3). The data collected with this camera will be used to develop machine-learning-based models for tumor tissue detection and to improve intraoperative navigation.

The equipment will be essential for conducting research activities, facilitating the development, testing, and validation of AI models, and ensuring their integration into clinical practice. The acquired infrastructure will allow efficient medical data processing, accelerate the development of innovative algorithms, and support the long-term sustainability of the research team within the ROBOMEDON project.

CVs of team members - complete according to the template provided; maximum of one A4 page per person. (See Guidelines for Applicants for the Call for Research Projects -2024)

1	***PERSONAL INFORMATION***	
	First name and last name:	Hrvoje Leventić
	Researcher's national ID:*	338691
	Website (ORCID and / or CROSBI):	https://www.croris.hr/osobe/profil/31452
2	***EDUCATION***	
	2019 dr. sc.	*Doctor of Technical Sciences - Joint PhD*
	Institution:	*J.J. Strossmayer University of Osijek, Faculty of Engineering, Computer Science and Information Technology Osijek, Ghent Unversity, Faculty of Engineering and Architecture*
	2012 mr. sc.	*magistar inženjer računarstva*
	Institution:	*Sveučilište J. J. Strossmayera u Osijeku, FERIT*
3	***EMPLOYMENT***	
	2013 – 2024 – position:	*Assistent, postodoctoral student, assistant professor*
	Institution:	*FERIT Osijek*
4	***LIST OF ACCOMPLISHMENTS*** *– the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).*	
	1. Leventić, H., Babin, D., Velicki, L., Devos, D., Galić, I., Zlokolica, V., Romić, K. & Pižurica, A. (2019) Left atrial appendage segmentation from 3D CCTA images for occluder placement procedure. Computers in biology and medicine, 104, 163 - 174 doi:10.1016/j.compbiomed.2018.11.006. [JCR Q1] [SJR Q1] IF=7.7	
	2. Habijan, M., Galić, I., Romić, K. & Leventić, H. (2022) AB-ResUNet+: Improving Multiple Cardiovascular Structure Segmentation from Computed Tomography Angiography Images. Applied Sciences, 12 (6), 3024, 18 doi:10.3390/app12063024. [JCR Q2] [SJR Q2] I F=2.7	
	3. Habijan, M., Galić, I., Leventić, H. & Romić, K. (2021) Whole Heart Segmentation Using 3D FM-Pre-ResNet Encoder–Decoder Based Architecture with Variational Autoencoder Regularization. Applied Sciences, 11 (9), 3912, 21 doi:10.3390/app11093912. [JCR Q2] [SJR Q2] IF=2.7	
	4. Romić, K., Galić, I., Leventić, H. & Habijan, M. (2021) Pedestrian Crosswalk Detection Using a Column and Row Structure Analysis in Assistance Systems for the Visually Impaired. Acta Polytechnica Hungarica, 18 (7), 25 -45 doi:10.12700/aph.18.7.2021.7.2. [JCR Q3] [SJR Q2] IF=1.711	
	5. Habijan, M., Babin, D., Galić, I., Leventić, H., Romić, K., Velicki, L. & Pižurica, A. (2020) Overview of the Whole Heart and Heart Chamber Segmentation Methods. Cardiovascular Engineering and Technology, 11 (6), 725-747, doi:10.1007/s13239-020-00494-8. [JCR Q4] [SJR Q3] IF=1.8	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	The "Vera Johanides" Young Scientist Award for the year 2021, awarded by the Croatian Academy of Engineering (HATZ)	
6	***MENTORING AND CO-MENTORING***	
	2023 – 2024 Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	*Mentor - 1 master thesis* *Co-mentor - 22 master thesis*
	Ustanova:	*FERIT Osijek*

1	***PERSONAL INFORMATION***	
	First name and last name:	*Marin Benčević*
	Researcher's national ID:*	*391271*
	Website (ORCID and / or CROSBI):	*https://orcid.org/0000-0003-4294-0781*
2	***EDUCATION***	
	2020. – 2024., dr. sc.	*Doctor of Technical Sciences - Joint PhD*
	Institution:	*Faculty of Electrical Engineering, Computer Science and Information Technology Osijek (FERIT Osijek), Faculty of Engineering and Architecture, Ghent University*
	2014. mag. ing. comp.	*Master of Engineering in Computer Science, Graduate University Study in Computer Science; Data and Information Sciences*
	Institution:	*FERIT Osijek*
3	***EMPLOYMENT***	
	2020. – present – position:	*Assistent*
	Institution:	*FERIT Osijek*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
	1. **Benčević, Marin**; Habijan, Marija; Galić, Irena, Babin, Danilo; Pizurica, Aleksandra, "*Understanding skin color bias in deep learning-based skin lesion segmentation*", *Computer methods and programs in biomedicine,* 245 (2024), 0; 108044, 10. doi: 10.1016/j.cmpb.2024.108044. ***Q1, IF: 6.1,*** Google Scholar citati: ***13***	
	2. Benčević, Marin; Galić, Irena; Habijan, Marija; Babin, Danilo; „*Training on Polar Image Transformations Improves Biomedical Image Segmentation*", *IEEE Access*, 9 (2021), 133365-133375. doi:10.1109/access.2021.3116265. ***Q1, IF: 3.557,*** Google Scholar citati: ***62***	
	3. Benčević, Marin; Qiu, Yuming; Galić, Irena; Pižurica, Aleksandra; „*Segment-then-Segment: Context-Preserving Crop-Based Segmentation for Large Biomedical Images*", *Sensors* (Basel). 2023 Jan 5;23(2):633. doi: 10.3390/s23020633.***Q2, IF: 3.557,*** Google Scholar citati: ***41***	
	4. Benčević, Marin; Habijan, Marija; Galić, Irena; Pizurica, Aleksandra, „Self-supervised Learning as a Means to Reduce the Need for Labeled Data in Medical Image Analysis",2022 30th European Signal Processing Conference (EUSIPCO) 1328-1332, doi:10.23919/EUSIPCO55093.2022.9909542. Google Scholar citati: ***13***	
	5.Benčević, Marin; Habijan, Marija; Galić, Irena; Babin, Danilo, „*Using the Polar Transform for Efficient Deep Learning-Based Aorta Segmentation in CTA Images*", *International Symposium on Electronics in Marine (ELMAR) 2022*, doi:10.1109/elmar55880.2022.9899786. Google Scholar citati: ***7***	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	2024 – Award for Excellence in Postgraduate Studies (FERIT Osijek), 2020 – Rector's Award, Josip Juraj Strossmayer University of Osijek	
6	***MENTORING AND CO-MENTORING***	
	2020 – present Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	*3 master thesis as co-mentor*
	Institution	*FERIT Osijek*

1	***PERSONAL INFORMATION***	
	First name and last name:	*Srđan Lazendić*
	Researcher's national ID:*	*Not applicable*
	Website (ORCID and / or CROSBI):	*https://orcid.org/0000-0003-2772-9240*
2	***EDUCATION***	
	2023., dr. sc.	*Doctor of Science, Mathematical Engineering*
	Institution:	*Ghent Universityu, Belgium*
	2016. msc theoretitcal mathematics	*Master of Theoretical Mathematics*
	Institution:	*University of Novi Sad, Serbia*
3	***EMPLOYMENT***	
	2023. – present – position:	*Doctoral Student and Assistant*
	Institution:	*Faculty of Engineering and Architecture, Ghent University, Belgium*
	2018. – 2023. – position:	*Doktorski student i asistent*
	Institution:	*Faculty of Engineering and Architecture, Ghent University, Belgium*
	2016. – 2018.– position:	*PhD Student and Researcher*
	Institution:	*Faculty of Engineering and Architecture, Ghent University, Belgium*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
	1. Robin Blommaert, **Srdan Lazendić**, Ljubica Oparnica, "The Euler-Bernoulli equation with distributional coefficients and forces" Computers & Mathematics with Applications, 123, 2022, 171 – 183, doi: 10.1016/j.camwa.2022.08.011, **Q1**, **IF: 2.9**	
	2. Zhao, T.; Montereale Gavazzi, G.; **Lazendić, S.**; Zhao, Y.; Pižurica, A. Acoustic Seafloor Classification Using the Weyl Transform of Multibeam Echosounder Backscatter Mosaic. Remote Sens. 2021, 13, 1760. https://doi.org/10.3390/rs13091760, **Q1**, **IF: 4.2**	
	3. Huysentruyt, R.; Van den Borre, I.; **Lazendić, S.**; Duquesne, K.; Van Oevelen, A.; Li, J.; Burssens, A.; Pižurica, A.; Audenaert, E. Sample Size Effect on Musculoskeletal Segmentation: How Low Can We Go? Electronics 2024, 13, 1870., **Q2**, **IF: 2.6**	
	4. N. Nadisic, Y. Arhant, N. Vyncke, S. Verplancke, **S. Lazendić**, A. Pižurica, "*A Deep Active Learning Framework for Crack Detection in Digital Images of Paintings*", *Procedia Structural Integrity* 64, 2173-2180 (2024), doi: 10.1016/j.prostr.2024.09.331. ***Q4, IF: 1.7***	
	5. **S. Lazendić**, J. Janssens, S. Huang and A. Pižurica, "On Interpretability of CNNs for Multimodal Medical Image Segmentation," 2022 30th European Signal Processing Conference (EUSIPCO), Belgrade, Serbia, 2022, pp. 1417-1421, doi: 10.23919/EUSIPCO55093.2022.9909776	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	2016 – 2018 – Sveučilište u Gentu, Belgija, UGent Bijzonder Onderzoeksfonds (BOF), grant br. BOF15/24J/078. 2018 – David Hestenes nagrada – za izvanredan rad i prezentaciju koju je održao mladi istraživač (mlađi od 35 godina), dodijeljena na AGACSE 2018, Campinas, Brazil	
6	***MENTORING AND CO-MENTORING***	
	2019 – present Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	*19 defended master's theses*
	Institution	*Faculty of Science and Faculty of Engineering and Architecture, Ghent University, Belgium*

1	***PERSONAL INFORMATION***	
	First name and last name:	doc. prim. dr. sc. Josip Samardžić
	Researcher's national ID:*	373440
	Website (ORCID and / or CROSBI):	*https://www.bib.irb.hr:8443/pregled/profil/35380*
2	***EDUCATION***	
	2013. – 2015. dr. sc.	*Dctor of Science*
	Institution:	*Faculty of Medicine, University of Tuzla*
	1985. – 1991. dr. med.	*Doctor of Medicine*
	Institution:	*Faculty of Medicine, University of Zagreb*
3	***EMPLOYMENT***	
	2016 – present – position:	*Director*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
	2020 – present – position:	*Assistant Professor*
	Institution:	*Faculty of Dental Medicine and Health, Department of Surgery*
	2019 – 2020 – present – position:	*Postdoctoral Researcher*
	Institution:	*Faculty of Dental Medicine and Health, Department of Surgery*
	2016 – 2019 – present – position:	*Assistant*
	Institution:	*Faculty of Medicine, Osijek*
	2005 – 2014 – present – position:	*Head of the Department of Surgical Diseases*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
	2000 – 2003 – present – position:	*Head of the Pediatric Surgery Department*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
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	2. GlobalSurg Collaborative and National Institute for Health Research Global Health Research Unit on Global Surgery. Global variation in postoperative mortality and complications after cancer surgery: a multicentre, prospective cohort study in 82 countries. Lancet 2021; 397 (10272):387-397.	
	3. GlobalSurg Collaborative and National Institute for Health Research Global Health Research Unit on Global Surgery. Effects of hospital facilities on patient outcomes after cancer surgery: an international, prospective, observational study. Lancet, 2022; 10, 7.	
	4. Mišković A, Pajić Matić I, Veselski K, Stojadinović T, Sauerborn D, **Samardžić J.**, Vučković I., Sinkroni planocelularni karcinom grkljana i intrahepatalni kolangiokarcinom – prikaz bolesnice, Med. Jad. 2022.	
	5. Vučković I., Pajić Matić I., Stojadinović T., Sauerborn D., Veselski K., Mišković A., **Samardžić J.**, Karcinom paratiroidne žlijezde – prikaz bolesnice., Med. Jad. 2023.	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	2020. *"Ladislav Rakovac" Award by the Croatian Medical Association for outstanding achievements in the development of healthcare, medical thought, and science, as well as for effective contributions to the Association. Memberships: Vice President of the Croatian Society for Endoscopic Surgery (2018), Vice President of the Croatian Society for Digestive Surgery (2015)*	
6	***MENTORING AND CO-MENTORING***	
	20? – 20? Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	-
	Institution:	-

1	***PERSONAL INFORMATION***	
	First name and last name:	Danijel Marinčić
	Researcher's national ID:*	-
	Website (ORCID and / or CROSBI):	-
2	***EDUCATION***	
	2025. - present	*Interdisciplinary Doctoral Study, Ruđer Bošković Institute*
	Institution:	*Ruđer Bošković Institute*
	2004. – 2010.	*Doctor of Medicine (M.D.)*
	Institution:	*Josip Juraj Strossmayer University of Osijek, Faculty of Medicine*
	2004. – 2010.	*Medical Technician*
	Institution:	*Medical High School, Slavonski Brod*
3	***EMPLOYMENT***	
	2022 – present – position:	*Head of the Operating Block and Central Sterilization*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
	2024 – present – position:	*Specialist Urologist*
	Institution:	*IMC Priora, Čepin*
	2021 – 2022 – position:	*Specialist Urologist*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
	2015 – 2021 – position:	*Specialist Urologist*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
	2012 – 2015 – position:	*Head of Emergency Medical Service Unit*
	Institution:	*Institute of Emergency Medicine of Brod-Posavina County*
	2011 – 2012 – position:	*Emergency Medical Team Leader*
	Institution:	*Institute of Emergency Medicine of Brod-Posavina County*
	2010 – 2011 – position:	*Medical Intern*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod*
	2011 – 2011 – position:	*Lecturer*
	Institution:	*Medical High School, Slavonski Brod*
4	***LIST OF ACCOMPLISHMENTS** – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).*	
	1.Conference UroOnko Forum, 24.10.2024. – 27.10.2024. Šibenik, Hrvatska	
	2.Intuitive DaVinci Multiport technology training program, Istambul, Hrvatska	
	3. IRE master class - Prostate focal therapy, Madrid, Španjolska	
	4. Urology today, predavač, Vodice, Hrvatska	
	5. Predavanje: „Robotic precison in prostatectomy: 400 Sehance extraperitoneal robotic radical prostatectomies“, Zagreb	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	European association of urology, Croatian Urological Society, Croatian Medical Association	
6	***MENTORING AND CO-MENTORING***	
	20? – 20? Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	-
	Institution:	-

1	***PERSONAL INFORMATION***	
	First name and last name:	Branimir Bošnjak, dr. med.
	Researcher's national ID:*	*Not applicable*
	Website (ORCID and / or CROSBI):	*Not applicable*
2	***EDUCATION***	
	2000. – 2009.	*Doctor of Medicine*
	Institution:	*Faculty of Medicine, Josip Juraj Strossmayer University of Osijek*
	1995. – 1999.	*Medical Technician*
	Institution:	*Medical School Osijek*
3	***EMPLOYMENT***	
	2015 – present – position:	*Urology*
	Institution:	*General Hospital Dr. Josip Benčević, Slavonski Brod / Private Hospital Priora*
	2012. – 2014. –position:	*Urology*
	Institution:	*Institute for Emergency Medicine of Brod-Posavina County*
	2012. – 2014. –position:	*Internship*
	Institution:	*Clinical Hospital Center Osijek*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
	1.	
	2.	
	3.	
	4.	
	5.	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	2004 – 2007: Scholarships and other forms of financial support for gifted students, Osijek-Baranja County. *2002 – 2004: National Scholarship from the Ministry of Science, Education, and Sports.*	
6	***MENTORING AND CO-MENTORING***	
	20? – 20? Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	-
	Institution:	-

1	***PERSONAL INFORMATION***	
	First name and last name:	Petar Nakić
	Researcher's national ID:*	39994
	Website (ORCID and / or CROSBI):	https://www.croris.hr/osobe/profil/39994
2	***EDUCATION***	
	2023 – present dr. sc.	*PhD Program in Electrical Engineering and Information Technology*
	Institution:	*University of Split, Faculty of Electrical Engineering, Mechanical Engineering, and Naval Architecture*
	2013. – 2015. mag. ing. comp.	*Master's Degree in Process Computing*
	Institution:	*University of Osijek, Faculty of Electrical Engineering*
	2010. – 2013. bacc. ing. comp.	*Bachelor's Degree in Computer Science*
	Institution:	*University of Osijek, Faculty of Electrical Engineering*
3	***EMPLOYMENT***	
	2020. – present – position:	*Lecturer*
	Institution:	*University of Slavonski Brod, Technical Department*
	2018. – 2020. – position:	*Lecturer*
	Institution:	*Polytechnic of Slavonski Brod*
	2015. – 2018. – position:	*Teacher*
	Institution:	*Vladimir Nazor Elementary School, Oprisavci*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
	1. Nakić, Petar; Katinić, Tomislav; Simić, Ranko; Marušić, Ivan, Perception of security risks in electronic data interchange of student population, Proceedings of the 10th International Scientific and Expert Conference / Damjanović, Darko ; Stojšić, Josip ; Mirosavljević, Krunoslav et al. (ur.). Slavonski Brod: Sveučilište u Slavonskom Brodu, 2022. str. 255-258	
	2. Nakić, Petar; Džambo, Igor; Lacković, Ivica; Ćosić, Tomislav, Application of computers and knowledge of ICT in agriculture, Proceedings of the 10th International Scientific and Expert Conference / Damjanović, Darko ; Stojšić, Josip ; Mirosavljević, Krunoslav et al. (ur.). Slavonski Brod: Sveučilište u Slavonskom Brodu, 2022. str. 393-396	
	3. „Međunarodni skup o prirodnom plinu, toplini i vodi", Sveučilište u Slavonskom Brodu, 2024.	
	4. 10th International Scientific and Expert Conference TEAM, 2022. Slavonski Brod	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	Institutional Projects: "Modeling Enterprise Innovation Using Intelligent Data Analytics" *Institutional Projects – "Green Economy"*	
6	***MENTORING AND CO-MENTORING***	
	20? – 20? Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	
	Institution:	

1	***PERSONAL INFORMATION***	
	First name and last name:	Maja Čuletić Čondrić
	Researcher's national ID:*	45964
	Website (ORCID and / or CROSBI):	https://www.croris.hr/osobe/profil/45964
2	***EDUCATION***	
	2022 – present dr. sc.	*Postgraduate Study*
	Ustanova:	*Faculty of Science, University of Split*
	2004. – 2009. dipl. ing. math	*Department of Mathematics*
	Institution:	*Faculty of Science, University of Zagreb*
3	***EMPLOYMENT***	
	2020. – present – position:	*Lecturer*
	Institution:	*University of Slavonski Brod, Technical Department*
	2016. – 2020. – position:	*Senior Lecturer*
	Institution:	*Polytechnic in Slavonski Brod*
	2012. – 2016. – position:	*Lecturer*
	Institution:	*Polytechnic in Slavonski Brod*
	2012. – 2012. – position:	*Assistant*
	Institution:	*Polytechnic in Slavonski Brod*
	2011. – 2011. – position:	*External Associate*
	Institution:	*Polytechnic in Slavonski Brod*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
	1. Novoselac, Vedran; Katić, Anita; Matijašević, Zlatko; Čuletić Čondrić, Maja, The Fixed Point Iterations Method for the Contraction Mapping, MM Science Journal, 2024, 5, 2024. str. 7613-7618. doi: 10.17973/mmsj.2024_11_2024086	
	2.Grizelj, Branko; Duspara, Miroslav; Stoić, Marija; Starčević, Valnea; Kokorić, Martina; Kokorić, Slaven; Čuletić Čondrić, Maja; Vuletić, Katica Calcution of bending position change, PLIN 2024 - Zbornik radova 15. međunarodnog skupa o prirodnom plinu, toplini i vodi, Raos, Pero; Kozak, Dražan; Raos, Marija et al. (ur.). Slavonski Brod: Sveučilište u Slavonskom Brodu, 2024. str. 209-218	
	3. Grizelj, Branko; Stoić, Marija; Starčević, Valnea; Čuletić Čondrić, Maja Bending moment, PLIN 2023 - ZBORNIK RADOVA 14. MEĐUNARODNOG SKUPA O PRIRODNOM PLINU, TOPLINI I VODI, Raos, Pero; Kozak, Dražan; Stojšić, Josip et al. (ur.). Slavonski Brod: Sveučilište u Slavonskom Brodu, 2023.	
	4.Čuletić Čondrić, Maja, Impact of worn plasma gun parts on cut quality, Proceedings of the 10th International Scientific and Expert Conference, Damjanović, Darko ; Stojšić, Josip ; Mirosavljević, Krunoslav et al. (ur.). Slavonski Brod: Sveučilište u Slavonskom Brodu, 2022. str. 299-305	
	5. Čuletić Čondrić, Maja; Stoić, Marija; Grizelj, Branko; Srbalj M. Exponential and logarithmic functions and their application in mechanical engineering // Proceedings of the 10th International Scientific and Expert Conference / Damjanović, Darko ; Stojšić, Josip ; Mirosavljević, Krunoslav et al. (ur.). Slavonski Brod: Sveučilište u Slavonskom Brodu, 2022. str. 347-353	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
	Institutional Project: "Research on Issues and Solutions in Technology, Processing, and Implementation in Educational Processes", Institutional Project: "Green Economy", Member of the Croatian Mathematical Society, Member of the Committee of the Professional Section of the Croatian Mathematical Society, Participation: Doctoral Joint Seminar Week, University of Ljubljana, Faculty of Education, Ljubljana, Slovenia	
6	***MENTORING AND CO-MENTORING***	
	20? – 20? Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows:	
	Institution:	

1	***PERSONAL INFORMATION***	
	First name and last name:	Nives Grčić
	Researcher's national ID:*	-
	Website (ORCID and / or CROSBI):	-
2	***EDUCATION***	
	2023 – present	*Graduate Study in Computer Science*
	Institution:	*Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
	2020. – 2023. bacc. ing. comp	*Undergraduate Study in Computer Science*
	Institution:	*Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
3	***EMPLOYMENT***	
	2023. – present – position:	*ETL Data Engineer*
	Institution:	*CROZ*
	2022. – 2022. – position:	*Student Laboratory Assistant*
	Institution:	*Faculty of Electrical Engineering, Computer Science, and Information Technology Osijek*
	2022. – 2022. – position:	*Software Development Intern*
	Institution:	*Vaco d.o.o.*
4	***LIST OF ACCOMPLISHMENTS – the list the five most important peer-reviewed publications or other works, such as articles in international scientific journals (with DOI number) or respositories (with links), patented research results, monographs, book chapters, conference papers and other research outputs (data, software, models, methods, theories, algorithms, protocols, art work and exhibitons, strategies, contributions to policy making etc).***	
5	*Other major academic accomplishments in the last 5 years (scholarships and awards; organisation of scientific conferences; memberships, etc.)*	
6	***MENTORING AND CO-MENTORING***	
	20? – 20? Number of theses defended / number of doctoral dissertations defended / number of post-doctoral fellows	
	Institution:	

Reference

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[3] H. D. L. H. a. C. J. C. Heibel, »A Review of Noninvasive Techniques for Skin Cancer Detection in Dermatology,« *American Journal of Clinical Dermatology,* pp. 1-12, 2020.

[4] H. S. C. Z. N. S. C. L. Riaz IB, »Applications of Artificial Intelligence in Prostate Cancer Care: A Path to Enhanced Efficiency and Outcomes,« *Am Soc Clin Oncol Educ Book.,* 2024.

[5] F. G. D. M. M. W. Betelhem Zewdu Wubineh, »Exploring the opportunities and challenges of implementing artificial intelligence in healthcare: A systematic literature review,« *Urologic Oncology: Seminars and Original Investigations,* svez. 42, br. 3, pp. 48-56, 2024.

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[7] I. H. a. J. W. H. Hwang, »A Study on Trends in AI Digital Textbooks and the Direction of Dedicated Font Design.,« *The Korean Society of Science & Art,* 2024.

[8] G. P. H.-I. Y. A. O. M. F. P. M.-M. B. Y. a. G. Ç. Sabatini, »Complete Digital Workflow for Fabricating an Occlusal Device Using Artificial Intelligence- Powered Design Software and Additive Manufacturing: A Dental Method,« *The International journal of prosthodontics,* svez. 37, br. 7, pp. 275-284, 2024.

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[19] R. M. C. L. X. Y. a. C. C. O. Lalmalani, »Artificial Intelligence in Dermatopathology: a systematic review,« *Clinical and experimental dermatology,* 2024.

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[34] B. L. Y. Z. P. H. N. Z.-X. L. a. Y. F. Wang, »MRI/TRUS-Guided Auxiliary System for Transperineal Prostate Biopsy Based on Deep Learning,« *2023 6th International Conference on Information Communication and Signal Processing (ICICSP),* pp. 914-921, 2023.

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